Dissertation Report

Economic and Monetary Policy Shocks in Indian Banking and Automobile Sectors, 2016–2024

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Undertaking

I, Abhimanyu Babbar, PRN 24021141003, have read the contents of the Dissertation Guidelines and have understood the implications and accept the terms and conditions mentioned in the Dissertation Guidelines.

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Authorization

I, **Abhimanyu Babbar (PRN: 24021141003)**, hereby submit my dissertation titled "**Economic and Monetary Policy Shocks in Indian Banking and Automobile Sectors, 2016–2024**", completed as part of the requirements for the Master of Business Administration (MBA – Finance) programme at Symbiosis Institute of Business Management, Hyderabad.

This dissertation has been prepared under the guidance of **Dr. Shyamsunder Chitta**. The work presented is my own and has been carried out during the course of my academic programme. To the best of my knowledge, this dissertation has not been submitted earlier, either in full or in part, to any other university or institution for the award of any degree or qualification.

All sources of information, data, and references used in this study have been appropriately acknowledged. I understand that the institute may make this dissertation available for academic reference and learning purposes, with due credit to the author.

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Date: 28/02/2026

Signature of student

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Abstract & Summary

Purpose of the Study

The research paper analyses that the impact of significant monetary policy changes and macroeconomic shocks on equity valuation multiple in India between 2016 and 2024. In particular, it explores how demonetisation, COVID-19 and extended monetary tightening affect Price-to-Earnings (P/E) and EV/EBITDA of banking and automobile companies. The study seeks to know whether the valuation reaction to these shocks varies among various industries that share the financial and demand characteristics.

Research Design and Methodology

The research design used in the study is a quantitative and longitudinal research design based on annual firm-level data of the major Indian banks and automobile companies between 2016 and 2024. The estimation of macroeconomic relationships between valuation multiples and macroeconomic variables is made using ordinary Least Squares (OLS) regression models. There are also interaction models that involve sectoral dummy variables to investigate the issue of asymmetric sectoral responding. It is based on an analysis using the asset-pricing theory, monetary transmission mechanisms, and valuation-multiple frameworks.

Key Findings

The results indicate that valuation responses to macroeconomic shocks differ significantly across valuation measures and sectors. Earnings-based valuation (P/E) is highly sensitive to periods of heightened uncertainty and prolonged monetary tightening, particularly in the banking sector. In contrast, enterprise-value-based valuation (EV/EBITDA) shows stronger sensitivity to changes in interest rates, especially for automobile firms. Interaction models reveal significant sectoral asymmetry, with banking firms experiencing sharper valuation compression during tightening phases, while automobile firms display greater sensitivity to financing cost changes.

Research Implications

A. Practical Implications:

The findings highlight the importance of using multiple valuation metrics when assessing firms under changing macroeconomic conditions. Investors and analysts should account for sector-

specific transmission channels when forming valuation and portfolio strategies, particularly during monetary tightening cycles.

B. Social Implications:

By revealing how monetary policy affects sectors differently, the study contributes to a better understanding of financial market stability and capital allocation efficiency. The findings may assist policymakers in evaluating the broader financial-market consequences of monetary interventions.

Originality and Contribution

This research contributes to existing literature by shifting focus from stock returns to valuation multiples, integrating multiple macroeconomic shocks within a single framework, and explicitly examining sectoral asymmetry in an emerging market context. The study offers novel insights into how valuation mechanisms differ across sectors during prolonged policy and economic disruptions in India.

Keywords

Monetary Policy Shocks, Valuation Multiples, Banking Sector, Automobile Sector, P/E Ratio, EV/EBITDA Ratio

1. Chapter 1. Introduction

1.1. Research Background

The 2016-2024 timeframe was a stormy period of the Indian economy. The restructuring of liquidity, credit behaviour, market expectations and sectoral performance were carried out through a series of policy shocks. As revealed by (Alessi & Kerssenfischer, 2019) monetary shocks impact on the prices of assets by modifying interest rates and change of expectations. Similar findings by (Banerjee et al., 2018) highlight the non-effectiveness of transparency in mortgaging and the creation of uncertainty in policy regimes on valuation behaviour.

In India, there occurred significant interventions that started with the demonetisation in 2016 that induced liquidity stress across cash-intensive sectors. Such liquidity shocks were reported to influence the market stability as stated by researchers such as (Anoop et al., 2018) and (Upadhyay & Suvarna, 2018). According to (Sinha et al., 2022), market adjustment was also shaped by the public sentiment. Following corporate financial activities such as LTROs and liquidity actions targeted market returns in the short term, which were in line with patterns reported by (Chundakkadan & Sasidharan, 2022).

Banking continued to play the centre stage in how the money was transmitted over this time. (Bhattacharya et al., 2008) indicate that the interest-rate fluctuations determine credit distribution and capital structure. (Amrutkar et al., 2022) have verified the existence of continued frictions in the pass-through mechanism of India. There is also transmission power of the market power and balance-sheet structures as suggested by (Coccorese, 2009). The announcements by policy redefinition valuation expectations at the banking firms, which is in line with (Marisetty & KM, 2025) findings.

The automobile industry reacted in some other way as it was pushed by the financing rates, family spending, and stability of the supply-chain. As per (Alexander & Al-Malkawi, 2023), the valuation of auto-sector is generally very sensitive to credit and macro signals. Disruptions resulting in the pandemic which (Behera et al., 2023) reported redefined the network of supply chains and production. Valuation dynamics were also affected by global rate cycles as (Forbes et al., 2024) suggest. The demand elasticity, financing, and reliance on imports contributed to the asymmetry in the reaction to valuation in the auto company, which is in line with the study by (Keswani et al., 2024).

Valuation theory provides the fact that stock price is based on the anticipated cash flows, and the discount rates. (CAMPBELL & SHILLER, 1988) and (Fama & French, 1993) discuss risk factors and expectations in the long-term, an aspect laid down in earlier work. (Easton, 2004), (Schueler, 2020), (Vishwanath, 2009), and (Damodaran, 2012) explain that valuation multiples use profitability, growth, risk, and macro conditions in structuring the multiples, such as P/E and EV/EBITDA. Shocks transform such valuation metrics to the extent it affects discount rates and earnings forecasts within industries.

The differing organizational form of the banking and automobile companies provides a possibility to study the response of the policy shocks to the adjustments of the valuation. The key mediums of response by banks are the interest channels, liquidity channels and the credit cycles. The automobile companies respond under the demand channels, the cost of financing terms and supply-side constraints. These disparities are the driving force to pursuant a comparative study of valuation results throughout a period of the most turbulent policies in India.

The 2016 to 2024 period of study was chosen deliberately to cover at least one monetary and macroeconomic cycle in India. This phase starts with the demonetisation shock of 2016, moves to the economic turmoil brought about by COVID-19 and finally enters the period of monetary tightening. Assessing this period allows the study of the valuation behaviour in structurally different regimes of policies, but maintaining the consistency of valuation metrics like P/E and EV/EBITDA, which are consistently available in the selected firms since 2016.

1.2. Problem Statement

Current Indian literature examines particular agents of change like the demonetisation, volatile pandemic or a single monetary announcement. But there is a dearth of literature that integrates these shocks into one and long horizon shock valuation framework lasting 2016-2024. Previous studies tend to concentrate on stock returns, stock volatility or transmission efficiency in contrast to the impact of macro shocks on valuation multiples. Relatives between banking and automobile companies are few even though their structure is essential and variously sensitized. This leaves a knowledge gap on the impact of major shocks on firm-level measures of valuation over a long policy cycle.

1.3. Research Objectives and Questions

A. Research Objectives

- To examine the effects of significant monetary and economic shocks in determining the valuation multiples of banking and automobile companies in 2016-2024.
- To make comparisons of valuation responses across demonetisation, pandemic upheavals and tightening periods.
- To determine sector specific avenues by which monetary shocks can affect the valuation outcomes.
- To test the changes in P/E and EV/EBITDA multiples at policy stages.
- To provide evidence on cross sector valuation sensitivity in Indian macroeconomic environment.

B. Research Questions

- What is the impact of demonetisation, pandemic disruption and interest-cycle cycles on multipliers of valuations?
- Are the banking and automobile companies haven't the same responses of valuation throughout policy periods?
- Which are the strongest transmission channels through which monetary affects the valuation outcomes on the different sectors?
- What are the characteristics of valuation dynamics within the periods of recovery and tightening cycles?
- What are the future value trends within the 2016-2024 period?

1.4. Significance of the Study

The research has implications to the academic research, financial practitioners, policy-makers and industry observers, as it offers a methodical analysis of the impact of grand macroeconomic and monetary shocks on company-specific valuation behaviour, in two structurally different India-based industries. Through the comparative study of the banking and auto industries between 2016 and 2024, a timeframe when the industry experienced demonetisation, the COVID-19 pandemic, and the ensuing tightening spiral, this study can be applied to the analysis of the key idea of how external shocks modify the valuation multiples, specifically the P/E and the EV/EBITDA. The results provide empirical insight into how monetary policy is transferred to equity valuation, a topic that gains more and more traction in the literature like

(Alessi & Kerssenfischer, 2019), (Anoop et al., 2018), and (Behera et al., 2023). It also informs us of sectoral asymmetry and how industries with varying financial endowments react differently in response to an identical set of macroeconomic forces which further adds into the discussion of cross-sector vulnerability and resiliency. Pragmatically, the research will assist investors, analysts, and portfolio managers to have improved knowledge on how the economic disruptions form the pricing behaviour, which will assist them to make better asset allocation and risk-managing decisions. To the policymaker, the results bring forth the impact that monetary interventions have on equity markets through the nontraditional avenues, which would aid in the more nuanced policy structuring in future. Accordingly, the study itself serves as a contribution to the current body of theoretical and practical research both in terms of bridging macro-financial dynamics with firm outcomes of valuation within a well-rounded, sector-specific context.

2. Chapter 2. Literature Review

2.1. Introduction to Literature Review

The behaviour of asset prices in the event of economic and monetary shocks has received rich scholarly interest especially in emerging markets where the policy mechanisms and institutional organisation are in a continual process of formation. Research by (Alessi & Kerssenfischer, 2019) notes that monetary shocks affect the valuation, based on the expectations, movements on the interests, and liquidity changes. A study by (Banerjee et al., 2018) also demonstrates the effect of policy-induced uncertainty on the quality of information used in markets in generation of price. These threads of effort contextualize the larger issue concerning the internalization of shocks between sectors with varying degrees of financial exposure, regulatory depth as well as demand sensitivity by markets.

Indian evidence is based on the global frameworks but displays the country-related transmission problems. The results of the analyses conducted by (Bhattacharya et al., 2008), (Amrutkar et al., 2022), and (Chundakkadan & Sasidharan, 2022) indicate that the rate pass-through and transmission of liquidity in the Indian system are different. The analyses of the auto industry and the banking business such as those by (Alexander & Al-Malkawi, 2023), (Klein & Machlica, 2021), and (Coccorese, 2009) indicate that the reaction to macro shocks is split because of the structure of a sector and dependence on financing sources.

This review of the literature brings together the world and Indian studies in the monetary policy, transmission channels, sectoral performance, valuation models, and structural disruptions to develop an overall framework of valuation dynamics in Indian banking and automobile companies in 2016-2024.

2.2. Monetary Policy Shocks and Asset Pricing

The interaction between monetary policy Shocks and Asset Pricing has been actively researched both in the global and emerging markets. Early studies by (CAMPBELL & SHILLER, 1988) form the basis that stock prices respond to changes in the expected dividends, discount rates and long-run expectations of the return, which suggests that any monetary shock changing the elements should result in an adjustment of the valuations. This mechanism has further been supported by studies conducted by (Fama & French, 1993) that indicated that common risk factors affect the stock and bond returns. These earliest models are the theoretical

grounds on which the unexpected changes in the rates of interest and liquidity conditions can be converted into the market movements.

Empirical data indicates that shock of monetary policy tends to have more impact on the prices of assets than expected. (Alessi & Kerssenfischer, 2019) illustrate that interest-rate surprises wreak disproportional changes in the prices of assets, mainly via expectations and risk-premium mechanisms. (Rigobon & Sack, 2004) report similar results, reporting that the announcement of the policy reflects into the equity valuations through both the interest-rate and the volatility channels. (BERNANKE & KUTTNER, 2005) make a substantial contribution to that literature by demonstrating that unforeseen instances of easing or tightening have instantaneous large stock market responses, indicating just how sensitive discount rates are to alteration of policy.

Policy shocks have different effects on markets in different countries in relation to institutional structures and market depth. Examples tell us that policy frameworks based on stable rules enhance macroeconomic stability and other premiums on uncertainty (Clarida et al., 2000). Nonlinear transmission effects and higher volatility are however common to the emerging markets. However, emerging markets often exhibit higher volatility and nonlinear transmission effects. (Bianchi et al., 2009) evidence suggest that policy credibility increases long-term interest rates stability, which is moderate in nature and shows the effect of improved monetary regimes. The knowledge gained highlights the need to examine the spread of shocks in country-specific situations like in India.

A number of works point to the industry-specificity of the impact of monetary policies. (Kim, 2023) observes that there are considerable equity adjustments in the emerging markets, even after the hikes in the U.S. rates and they spill over to domestic sectors with different intensities. (Gode & Ohlson, 2000) also demonstrate that valuation multiples will shrink when interest rates are increasing as under such circumstances, the present value of expected cash flows falls with an increase in discount rates. (JARROW & TURNBULL, 1995) work on credit-underlying-Pricing supports the view, that under monetary shocks risk profiles of firms vary and affect pricing of assets in an industry difference.

On the whole, according to the literature, monetary shocks impact in many ways on the prices of assets that include discount-rate changes, liquidity, expectations, and risk premiums. Nonetheless, the majority of studies in the world will present aggregate market level evidence,

and not industry specific understanding as no much detail can be given on how structurally diverse industries will react to long term shocks. This is a void that drives the necessity in specific sectoral assessment specific to an Indian banking and automobile sector.

2.3. Monetary Transmission in India

The issue of monetary transmission in India has received significant academic interest because there is forever the concern over its effectiveness, temporality, and heterogeneity of the sectors. Applied data on interest-rates as practiced by (Bhattacharya et al., 2008) indicates that interest-rates cycles in India often fluctuate with a significant number of lags, and as such the cycles reflect structural rigidities present in the financial system. These authors argue that the banking channel of transmission is hampered by interest rates that are administered, imperfect competition, and balance-sheet incompatibilities. Their results suggest that the interest-rate changes caused by policies do not follow a consistent pattern in terms of their transformation in lending or deposit rates hence creating a randomized environment to the firm that depends on external funding.

The structures of inefficiencies prevail, even though there has been an evolution in the transmission dynamics. (Amrutkar et al., 2022) explore the concept of interest-rate pass-through in India and emerge with the result that the change in policy rates leads to asymmetric effects on the whole financial system. The lending rates have a less responsive adjustment and deposit rates are more immediate in response, which introduces frictions in the price of credit. These frictions affect the valuation of rate sensitive firms especially Banks and those heavily levered. The authors drive home the point that the interest-rate channels are very much dominant as the transmission channels with the liquidity channels playing a considerable role during economic stresses.

The liquidity-oriented interventions have become increasingly determinant in the context of developing the Indian environment of transmission. (Chundakkadan & Sasidharan, 2022) assess the effect of long-term operations of the repo lien (LTRO), achieved through using the equity returns, and report significant market responses after repaying the liquidity. Their findings indicate that the liquidity channel has become more significant in the post 2016 years, particularly after the Reserve bank of India implemented unconventional methods of stabilizing markets. The dependency on liquidity activities reveals that credit market is vulnerable to the

current conditions of short-term funding directly influencing valuation product of industries that are susceptible to credit cycles.

The Indian transmission further refers to sectorial differences which complicate the transmission landscape. According to (Ghosh, 2019) the degree of monetary transmission between industries is hugely different, and interest-rate shocks also have strong impacts on capital-intensive and cyclical industries. The paper notes that sectors that are highly dependent on financing show a higher response to policy changes, which makes it imperative to analyse the sector in question. Similarly, (Keswani et al., 2024) record long-term co-integration association between macroeconomic variables and stock prices in India, thus supporting the idea that the transmission patterns across industries vary according to structural peculiarities.

Asymmetric transmission patterns were increased by shocks like demonetisation and the pandemic. According to (Anoop et al., 2018) the impact of the demonetisation episode on different industries was imbalanced and partially caused by liquidity withdrawals and changing credit conditions. In the COVID-19 crisis, (Behera et al., 2023) show how uncertainty, liquidity stress, and policy responses brought in different transmission effects of both financial and non-financial sectors. These results demonstrate that the extraordinary shocks are more likely to amplify the sectoral differences in the effects of the monetary impulses on the price of assets.

There are also international spillovers that are imposed on the transmission framework of India. According to (Kim, 2023), U.S. policy tightening triggers massive equity market responses in the emerging economies such as India via risk-premium and expectations. Domestic transmission, therefore, should be closely interpreted in relation to international cycles of rates and especially in areas where foreign investments have been made or they have exposure to imports.

Although there is a mass of literature suggesting this, there remain gaps in how transmission of monetary operations impacts valuation multiples as opposed to just stock returns or volatility. Indian research mostly shows short time market adjustments, or macro pass-through but shows little evidence of the behaviour of firm-level valuation measures over long shock durations. Furthermore, comparisons across sectors are still very limited particularly in banking sector as well as the automobile sector which also vary greatly in terms of financing structures, demand conditions as well as being sensitive to balance sheets. Such constraints justify the

importance of a sector-specific valuation analysis that covers the 20162024 period, with several financial and economic shocks.

2.4. Banking Sector Evidence

Banking industry is placed at a pivotal point of transmission of money, and the valuation relationships in banks are closely connected with the policy-related changes in interest rates, liquidity status, and credit risk. (Coccorese, 2009) notes that the pricing behaviour and responsiveness of the banks to monetary impulses is highly dependent on the market power. Banks that have a greater concentration in the market have a great pricing power making them to have asymmetric adjustments in lending rates and deposit rates. These institutional characteristics influence the manner in which policy shocks are converted to profitability, legalization as well as eventually valuation multiples. The results suggest that money shocks not only affect bank valuations by having an effect on the discount-rate, but also via competition factors within the industry.

There exists a body of literature that studies the behaviour of banking-valuations to changes in interest rates. (Gode & Ohlson, 2000), illustrate that multiple of P/E condenses when the level of interest rates increases indicating increase in discount rates and reduced expectations of cash flows in the future. Even though they do not specifically refer to India in their model, they provide a theoretical basis to the rate sensitivity in Indian banks. In their research on universal banks in the Philippines, (Dayag & Trinidad, 2019), conclude that the driving factors of valuation, including profitability, risk positioning, and growth prospects are significantly affected by macroeconomic factors. Their implications may be used by Indian banks despite having similar regulatory structures and because the market has been formed in a similar way although based on another emerging market.

Evidence in the Indian-sector shows that there is a strong valuation adjustment during macroeconomic disruptions. (Anoop et al., 2018) reveal that there was a positive short-term effect on the banks due to demonetisation as a result of an unusual rise in deposits and a short-term increase in liquidity. Their findings show that external shocks can build the balance sheet of banks in the short run despite the larger economic pressure. Similarly, (Upadhyay & Suvarna, 2018) note the existence of high volatility in the banking stocks in the times of demonetisation, which can be explained by changing the state of liquidity and the changes in

the investor sentiment. These papers support the idea that liquidity shocks are instantaneous effects on the bank valuations, despite the expansion of the wider economy that is contracting.

The COVID-19 increased the pressures related to valuation as it provided credit risk and provisioning issues. India is among the examples of monetary policy impact added by uncertainty brought about by the pandemic (Behera et al., 2023). Their findings indicate that banking industries are also more sensitive to the liquidity and announcement of policies during crisis times. Corresponding to this view, (Mohanty & Mishra, 2024) note that the market of India underwent powerful overreaction during the pandemic, and banks demonstrated the quickness of price adjustments due to the sense of uncertainty and aversion to risk. These researchers emphasize that unusual shocks do not just change the normal transmission environment but also, they do not have the same valuation processes as usual policy cycles.

Long-term interest-rate behaviour variations are the root causes of banking valuations. (Bianchi et al., 2009) argue that increased monetary credibility will have the effect of stabilizing the long-term rates and reducing uncertainty premiums, which will have the potential effect of supporting higher valuation of rate-sensitive institutions. Similar forces at work in the Indian context are observed at times of virtually stable policies or predictable rates cycles. Moreover, (Forbes et al., 2024) note that international rates impact the national financial environment, which is an important fact to take into account among Indian banks, which are open to the international investments and exposure to the external sentiment of risk.

Evidence in the form of events of India supports the increased sensitivity of banks to announcements. As evidenced by (Marisetty & KM, 2025), interest-rate announcements cause dissimilar reactions in banking and non-banking firms whereby banks have more significant reactions attributed to the direct exposure to the policy-directed changes in the cost of funds and credit terms. These observations lend more weight to the debate on whether bank valuations need to be studied within the barriers of monetary and economic shocks, as opposed to subordinating them under a homogenous system of the equity market.

Despite having a rich literature, there are still some gaps. Stock returns, volatility or short-term response has been predominately focused on, and the relatively small emphasis on valuation multiples (price-to-earnings (P/E) and enterprise value to earnings before interest, taxes, depreciation, and amortisation (EV/ EBITDA)) has been given. Furthermore, there has been no comprehensive study of the long-term valuation trends of the banking industry in a series

of shocks, such as demonetisation, the pandemic, and tightening cycles, in the Indian context. The absence of a sustained, multi-phase valuation analysis in such a way brings out the need to have a comparative framework, on the basis of combining monetary policy, transmission frictions, and sectoral fundamentals to explain how banks re-priced risk in the period 2016-2024.

2.5. Automobile Sector Evidence

The Indian economy places the auto industry at a unique position because the company is dependent on demand in households, availability of credit, stability in the supply-chain and prices of inputs globally. Being consumer-led and capital intensive in nature, the automobile companies are generally more sensitive to monetary cycles, financing terms as well as macroeconomic fluctuations. (Alexander & Al-Malkawi, 2023) emphasize that the effect of macroeconomic variables on the Nifty Auto Index is sharp, especially when it comes to interest rates, inflation, and industrial production. Their results indicate that the automobile values are affected by variation in the cost of borrowing nearly immediately, indicating the cost borne by the consumers in financing a vehicle and the manufacturers in maintaining a working capital.

Valuation behaviour is also determined by structural factors. In the study by (Klein & Machlica, 2021), the authors discuss how the pandemic of the COVID-19 crisis impacted automotive companies in Central and Eastern Europe and discovered that disruptions in supply chains, labour shortage, and a weakened demand intensified the sensitive to macroeconomic shocks. Even though their observations are made in the European setting, they are representative of trends in India where lockdowns constituted production schedules and inventory bottlenecks. The (Sarin, 2024) complementary evidence points out financial stress and unstable production cycles as a long-term cause of the up-and-down movements of the valuation in modern auto market.

The studies on the valuation at the firm-level also highlight the significance of the financial structure and operational efficiency. In her comparative valuation study of automobile industry (Liu, 2019) arrives at a conclusion that the valuation multiples are not only dependent on the profitability, but also on the intangible data (brand strength, market share, technology adoption, etc.). The paper shows that macroeconomic shocks would change the valuation results by changing the investor expectations concerning the competitive nature over the long term. Likewise, (Kumar, 2025) uses predictive modelling to demonstrate that stock prices in the

automobile sector are highly sensitive to macro indicators such as growth of GDP and interest rates, i.e. towards which multiples of valuation have implications of future expectations of cyclical recovery.

The sensitivity of the sector to financing costs is quite pertinent especially to India, since the portion of purchases of cars through bank credit or non-bank financial companies (NBFCs) is too big. A literature on oil and commodity correlations like (A. Jain & Biswal, 2016), shows that exchange-rate fluctuations, volatility in crude-oil prices, and supply trends throughout the world affect production cost and valuation patterns. These variables interplay with shocks in the monetary policy and enhance the fluctuations of the valuation, as the movement of the costs of taking loans and the prices of inputs coincide.

There was also a high disruption in demands due to the demonetisation episode. (Anoop et al., 2018) list the significant decrease in consumer expenditure in cash-sensitive sectors, such as automobiles, in the demonetization process. Even though their work is general and looks at the performance of the sectors in general terms, the implication to the auto manufacturers is evident: liquidity withdrawal is influential as it impacted negatively the retail demand and stressed the valuation. The same is observed by (Upadhyay & Suvarna, 2018), who note the volatility due to panic and short-run drops of auto related stocks during the event.

The strength of the sector was increased due to pandemic disruptions. (Behera et al., 2023) demonstrate that the transmission of the monetary policy had been greatly changed because of the uncertainty that came along with pandemic, among those sectors mostly exposed to supply-chain and consumer-demand shocks. Their results suggest that auto companies, which were much sensitive to both the nature of demand and the international input, were severely volatile with regard to their valuation. (Mohanty & Mishra, 2024) also observe that there was an over-reaction of the Indian market to the pandemic, especially in sectors in which revenue visibility failed suddenly, which is also rising in line with the performance of the auto-industry during lockdown waves.

Multi-year tests explain long-term susceptibility to the business end-of-policy cycles. According to the results presented by (Keswani et al., 2024), cointegrated relations are observed between macroeconomic variables and stock prices; also, auto companies exhibit more robust and consistent associations in comparison with financial-sector stocks. This implies that an increased number of macroeconomic expectations (such as industrial output,

inflation, and interest rates) is priced into automobile valuations, and thus they are more susceptible to cyclical booms and busts.

Despite the large quantity of literature on auto-sector valuation, there are still a number of limitations. Numerous studies use sales performance, price responses, or macro associations, but few of them analyse the valuation multiples including P/E or EV/EBITDA over longer periods of shock. In addition, comparative studies across sectors relating monetary policy shocks to the outcomes of valuation of both the banking and the automobile firms are also limited. Such a gap highlights the necessity to consider the valuation theory, the workings of monetary transmission, and the sector features to analyse how auto companies re-priced the risk during the major Indian shock incidents between 2016 and 2024.

2.6. Valuation Models and Multiples

The analytical basis of valuation models and multiples entail the establishment of a pricing mechanism of firms in different macroeconomic and monetary settings. Within the traditional valuation theory, firm value is the present value of discounted future cash flows of the firm based on the risk and growth prospects as well as macroeconomic factors. The seminal article of (CAMPBELL & SHILLER, 1988) outlines the basic attention to which the expectation of dividends and discounts and long-term returns of stocks are connected and contains a justification of the fact that the values indicators are highly sensitive to any changes in the monetary policy and the macroeconomic outlook. It is based on this that (Fama & French, 1993) are able to show that systematic risk factors such as market risk, size, and value characteristics typically dominate the equity returns and valuation behaviour across the industries.

Various valuation multiples are very common especially the P/E and EV/EBITDA due to their simplicity and comparability. The relationships conceptualised in multiples (Damodaran, 2012) demonstrates that P/E ratios are the result of growth expectations, payout policies, and required returns. His model points to the fact that though tiny shifts in discount rates, which are usually caused by the shock in money, can produce tremendous impetuses. This argument is supplemented by (Easton, 2004) which studies the application of the P/E ratios in estimating the implied expected rates of return; a high required return corresponds to tight monetary policy, which tends to squeeze the valuation multiples. The conceptualization of the

relationship between macroeconomic variables into market-based measures of valuation is taken in these works.

More explanation of the characteristics of multiples is offered by (Schueler, 2020), who claims that valuation multiples no longer should be studied outside the frame of the consistent economic and accounting domain. He emphasizes the fact that the stability of multiples is determined by stability of earnings, cash flows and capital structures- things that are often unstable when there is an economic hit. In the same vein, (Vishwanath, 2009) describes that multiples can contain data on the profitability, leverage and risk-adjusted growth meaning that when macroeconomic uncertainty arises or money tightens it can change the valuation in firms of the same type with no changes in their fundamental aspects. These theoretical concepts are supported by evidence in emerging markets. According to (Ramcharran, 2002) the P/E ratios of developing economies respond largely to the macroeconomics variables such as interest rates, inflation rates as well as exchange-rate fluctuations.

A number of the empirical studies associate monetary shocks with multiple-based valuation. Indeed, (Gode & Ohlson, 2000) demonstrate that an increase in the interest rates leads to lowering of P/E in that they accelerate the discount rates and minimizes the value of the future earnings. Their results are in line with the rest of the asset pricing literature, that predicts numerous contractions during periods of tightening. Moreover, PEG ratio centrality (Chan, 2023) offers a more naturalized methodology of the equation, which supports the idea that valuation values reflect expansion in comparison to the cost and that the multiples can be altered significantly through monetary shocks that alter expectations of expansion. These findings can be used to propose the hypothesis that valuation measures are a sensitive measure of market internalisation of policy change.

Cross country analyses also indicate the valuation model responses to various macroeconomic environments. According to (Bianchi et al., 2009) the term structure of interest is significantly moderated after credible policy regimes by the monetary authorities implying that the more the monetary policy is credible the lower the level of uncertainty premium and the higher valuation multiples. The same is noted by (Forbes et al., 2024), who make the assumption that rate cycles in the world will affect the domestic risk premiums in the emerging markets, thus, influencing the sectoral valuations.

Firm specific attributes are also constituent of the impact of macro shocks to valuation multiples. At the same time, (Dayag & Trinidad, 2019) discover that the drivers of bank valuation are profitability and capital adequacy, and vulnerability to risk, but that macroeconomic factors are also present in terms of discounts-rate. The Chinese evidence presented supporting (Gupta, 2018) reports the results of Indian companies and confirms that profits, leverage, and growth perspective of the companies drive the valuation multiples, and the investors include firm specific and macroeconomic information asymmetrically in cross sectoral settings. Her results indicate that there should be sector specific analysis of valuation especially when the macro shocks are changing the discounts, the liquidity movements or the earnings visibility.

Nevertheless, there exist gaps in the general valuation literature. Much of the available literature studies behaviour of valuation multiples in stable macroeconomic conditions which provide minimal evidences on how such measures change over time durations in case of prolonged shock periods especially in emerging markets. There are almost no research imbibing demonetisation, pandemic disruptiveness, and tightening cycles into a single, synthesised valuation framework. Moreover, the comparisons across industries, particularly banking and automobile companies are underexplored, although these industries have opposite funding models, and are sensitive to bank policy. These loopholes explain why a long term, event-based valuation of the 2016-2024 period is necessary.

2.7. Structural Breaks and COVID-19 Evidence

The structural breaks are discontinuities in the economic relationships usually due to significant shocks that change expectations, liquidity or risk perceptions. India has a number of such breaks in the period 2016-24, namely: demonetisation, the COVID-19 pandemic, and the monetary tightening cycle, all of them have changed the way markets operate and the market sector perform. The literature on structural changes points out, in the economic time series, that the behaviour of the time series varies in different regimes, and that failure to take this variation into account creates bias interpretations of valuation changes.

The example of emerging economies shows that the process of structural transitions is complicated. (Tomar & Kesharwani, 2022) discover unequal impacts of monetary policy on Indian sectoral returns with tightening events producing steeper repercussions than loosening events. The results of their investigations point to nonlinear dynamics, and indicate that shocks

cause regime specific valuation effects. In line with this view, (Sahoo, 2021) records severe structural discontinuities in Indian stock markets amid the COVID-19, and the traditional transmission connections in the more uncertain scenario were also found to be more fragile. These findings suggest that regimes change, and valuation models should have the ability to capture change and not only regimes that react consistently (with comparable responses over time).

The demonetisation shock created a structural break of a kind in the period that we examine. (Anoop et al., 2018) demonstrate that demonetisation caused a rapid volatility of the sector because of a sudden loss of liquidity. Their data shows how they had different effects: banks initially experienced the positive effect of inflows in the form of deposits whereas industries that are dependent on cash-like industries such as automobile and consumer discretionary sectors performed worse. In line with this assessment (Upadhyay & Suvarna, 2018) determine that market sentiment and volatility patterns shifted upon immediate demonetisation to the extent that they implied that investors reviewed the risk and earnings expectations as a reaction to the immediate liquidity stress.

The COVID -19 crisis is the largest structural disruption of the decade. (Behera et al., 2023) establish that monetary policy effects were increasing in emerging markets in Asia including India due to the uncertainty surrounding the pandemic. Their analysis proves that the policy announcements in the time of crisis had an increased effect on the market, and more successful responses were found in those sectors which are responsive to the liquidity and demand conditions. On the same note, (Mohanty & Mishra, 2024) depict the Indian markets as having an over-reaction to the development of COVID-19 representing an extreme level of uncertainty and short-term pessimism. According to the authors, the severity of the structural break itself manifested in the price dislocations as well as in valuation adjustments, especially in metric industries which are vulnerable to both, supply-chain shocks and price shocks.

The existence of structural breaks in Indian industries is supported by labour and production dynamics evidence. According to (Mishra et al., 2024) , structural adjustments take place over time in sectoral changes that have been initiated by liberalisation reforms of 1991 and further macro disruption thereafter. The results of their study show that extreme shocks may have enduring changes on sectoral pathways, labour allocation and production nexus. These pieces of information clarify why COVID19 caused more permanent side effects in car manufacturing

and inventory turnover, and banking valuation was informed more by the perception of credit pressure and the need to make provisions.

Structural changes got enhanced due to financial contagion and global spillovers. According to the research by (Kim, 2023) and (Forbes et al., 2024), the emerging markets have exaggerated reactions to international uncertainty and global rate fluctuations. These ties were further strengthened by COVID -19 that triggered more volatility globally and tightened the financial conditions. In the context of Indian, these spillovers were both in the banking and automobile industries by influencing funding conditions, exchange rate, and global supply chain changes.

In spite of the increased literature, there are gaps that exist. Structural breaks in returns or volatility are studied by most studies as opposed to valuation multiples. Not many of them combine demonetisation, COVID 19, and tightening episodes into a single framework. In addition to that, the comparative evidence available on the repricing of risk by banking and automobile companies under different structural regimes between 2016 and 2024 is limited. These exclusions reinforce the evidence of a holistic sector-level analysis of valuation, whereby structural changes, event periods, and macroeconomic changes are factored into place.

2.8. Cross-Sector Comparative Insights

The cross-sector analysis can be highly useful in the identification of different industries to understand ways in which industries with varying structural, financial, and operational characteristics react to monetary and macroeconomic shocks. Banking and automobile industries used in India are two very different spheres: the former is well incorporated in credit and liquidity networks, whereas the latter is guided by market demand, production, and the state of financing. These structural variations determine non-homogenous valuation provisions whenever the economy determines disruption, interest-wave shocks or liquidity transformations.

According to a number of studies, it is important to understand mechanisms which are specific to sectors. (Ghosh, 2019) shows that the monetary policy transmission is higher in India depending on an industry, and capital-intensive and credit-sensitive industries are those that are more sensitive to policy changes. According to these findings, there are macro shock absorption mechanisms unique to industries associated with the financing cycles, including automobiles and banks. In the same manner (Keswani et al., 2024) indicate that stock prices in the Indian industries demonstrate long-term effects of macroeconomic variables in the industry,

even though the strength and direction of partaking differ between industries. Their findings suggest that sectoral fundamentals, as well as cyclical sensitivities affect valuation multiples not just based on domestic monetary cues.

The significance of cross sector differentiation is supported by international evidence. (Forbes et al., 2024) form the opinion that there are disproportionate global rate cycles culminating in the parts of the economy which are interested in interest; the parts are those whose values change more significantly. Their evidence indicates that banking and automobile companies which are extremely affected by the state of finance might react differently to the balance-sheet exposures, product demands, and various merits of the supply-side and credit-side channels. Further supporting this cross-country understanding is that (Ouyang & Rajan, 2013) consider exchange-rate fluctuations of the lead up to what they reveal of a non-tradable intensive sector to respond diversely to a macro shock relative to the tradable industries. This difference fits with the Indian situation in which automotive companies rely largely on the imported parts and banks work in the domestic legal and financial frameworks and settings.

Sector studies also serve to support asymmetry. As (Anoop et al., 2018) demonstrate, the process of demonetisation was unequally distributed in the economy, with the banks advantageous due to the inflow of liquidity and the auto firms negatively with the shrinking demand. Equally important, the volatility observed by (Upadhyay & Suvarna, 2018) in the process of demonetising highlights the fact that the sectors that were cash-intensive were adversely affected by disruptions in the short-term valuation. These results point to the fact that the same macro shocks may have the contrary valuation impact on sectors based on their dependence on cash transactions, liquidity statuses, and credit intensity.

The pandemic of COVID-19 has shown clear sectoral inequality. (Behera et al., 2023) establish that the impact of monetary policy in emerging markets was increased by the uncertainty brought about by the pandemic, and sectors that are given high liquidity sensitivity are more sensitive. The automotive makers had to face supply-chain implementations, manufacture halts and demand retractions but the banks had to experience measurement of credit risks and the need to provision. (Mohanty & Mishra, 2024) also note that Indian markets have a strong response to the pandemic, which is an indicator of the differentiated classification of sectors and nonhomogeneous recovery expectations. Their results highlight that the multiples of valuation are not only sensitive to the current earnings, but also sensitive to the perceived

stability of future cash flows, which broke away significantly between banking and auto companies throughout the crisis period.

It has also discussed the sectoral performance in line with interest-rate cycles. According to (Tomar & Kesharwani, 2022) , adding to the monetary policy transforms asymmetric effects in industries, but stricter monetary policy negatively affects companies, which depend heavily on external capital. This forms a direct connection between automobile valuations and cost-of-credit dynamics, and banking valuations are largely determined by which they expect the margin to be and which credit of what quality. Similarly, (N. Jain, 2025) provides evidence that the Indian stock market is characterized by a strong actual reaction to changes in reputation rates, but sensitivity to the constraints of the discount-rate is unevenly distributed across sectors so that the extent of the reaction differs across sectors.

Starving as is these contributions; little comparative valuation research has been done. There are limited studies that have combined more than two shocks, which are the demonetisation, COVID-19, and tightening cycles, within one cross-sector analytical framework. The vast majority of studies concentrate on price responses or the connection between the macroeconomy and not on valuation multiples including price-to-earnings and EV-EBITDA. Additionally, there is little empirical research that compares industry segments that differ substantially over long policy processes as closely as banking and automotive. These lapses indicate a need to carry out a thorough long-run study that review the way the two industries internalise macroeconomic shocks with valuation adjustments, which improves our comprehension of sector asymmetry in emerging-market situation.

2.9. Identification of Research Gaps

An analysis of the existing literature reveals a number of limitations that require an in-depth, sector-specific study of the valuation between 2016-2024. As numerous studies focus on monetary policy shocks and the price of assets, most of them, such as by (Alessi & Kerssenfischer, 2019), (BERNANKE & KUTTNER, 2005), and (Rigobon & Sack, 2004), focus on a response at the aggregate market or aggregated asset prices. These papers provide information on the effect of interest-rate shocks on market prices but do not give industry specifications as to the explanations of how valuations change across industries with substantial diversity in financial policy, and macroeconomic responsiveness.

On the Indian setting, the research works by (Bhattacharya et al., 2008), (Amrutkar et al., 2022), and (Chundakkadan & Sasidharan, 2022) examine the monetary transmission channels and identify structural frictions in the policy pass-through. Nevertheless, these analyses are more on rate adjustments, liquidity flows and credit allocation as opposed to one at a firm level of valuation. Therefore, the overall issue of the long-term relationship between monetary shocks and multipliers of value is poorly studied.

Limitation is further illuminated through sector alike research. Studies that address the banking sector. Works by (Coccorese, 2009), (Dayag & Trinidad, 2019), and (Marisetty & KM, 2025) provide the insights into the price power, determinants of valuation, and the sensitivity of policies to the analysis, but do not investigate how valuation multiples, such as price-to-earnings or EV/EBITDA move as a result of successive macro shocks. The changes in the macroeconomy, demand conditions are recorded in literature of the automobile-sector such as (Alexander & Al-Malkawi, 2023), (Klein & Machlica, 2021), (Liu, 2019), and (Sarin, 2024); nevertheless, the studies primarily focus on short-term price changes and production-related problems or predictive modelling instead of shock-induced valuation changes over the longer-term horizons.

Both structural discontinuities and outstanding shocks add to the scenery. The works of (Anoop et al., 2018), (Upadhyay & Suvarna, 2018), (Behera et al., 2023), and (Mohanty & Mishra, 2024) prove that regime changes and increased volatility are caused by such events as demonetisation and COVID-19. However, the analyses are limited to price reactions, volatility clustering, or uncertainty effects, but not to valuation multiples hence creating a gap in understanding how the long-term firm valuation varies across different macroeconomic regimes.

There is little cross-sector comparative information. Although (Ghosh, 2019), (Keswani et al., 2024), and (Forbes et al., 2024) find that sector-specificities in reaction to macro variables exist, their studies do not involve recurrence of multiple shocks and do not draw up valuation behaviour among banking and automobile companies. The available literature tends to conduct individual sector analysis, overlooking the fact that such industries have opposing financial patterns, patterns of demand, and exposure modes.

There are three notable gaps identified in the wider literature. To start with, there are insufficient studies that make an evaluation of the valuation multiples, but not stock prices

during the times of significant macroeconomic and monetary shocks. Second, the three major Indian shocks (demonetisation, COVID-19, and tightening cycles) have rarely been placed in one analytical framework in existing scholarship. Third, there is a conspicuous absence of long-horizon cross-sector comparative valuation studies that can compare the banking industry dominated by credit, liquidity and regulation factors, with the automobile sector dominated by demand cycles, financing terms and supply-chain relationships.

These gaps explain the need to have a longer, shock-driven, sectoral valuation study that looks at 2016-2024 that enables us to look systematically at the changes in valuation multiples of structurally different sectors in India due to monetary and economic shocks. Moreover, Indian research that exists rarely explicitly conducts statistical comparisons across sectors based on valuation multiples, and thus offers an opportunity to conduct cross-sector inferential testing.

2.10. Theoretical Framework

The conceptual framework of this study is based on monetary economics, asset-pricing theory and valuation models that explain how the macroeconomic factors influence decision-making at the firm level in terms of pricing. The main part of the discussion is based on the principle of discounting since it assumes that the value of an asset is the same as the present value of future cash flows of the asset (CAMPBELL & SHILLER, 1988). The authors promote a long-horizon model which is used to associate the stock prices with the potential dividends, expected rate of discount, and future returns, thus showing that valuation ratios like the price-earnings (P/E) multiple react to the changes of monetary policy which may alter either the discount rate or the expectations about future profitability. Likewise, the factor-based model that has been explained by (Fama & French, 1993) suggests systematic risk factors that define change in portfolio returns, implying that a change in the microeconomic conditions can remodel the sector risk patterns and, therefore, predict changes in the valuation results.

The rules of monetary policy are also another pillar in the theoretical approach. (Clarida et al., 2000) prove that the predictable, rule-based policy structures bring stability in inflation and narrow the uncertainty premiums, and discretionary or unpredictable interventions trigger exaggerated asset-price responses. This analysis on their part suggests valuation multiples are expected to respond to predicted and unexpected monetary interventions. To complement, (Rigobon & Sack, 2004) show that monetary shocks do spread the equity prices both over interest-rate channels and through the transition of market volatility and risk-taking. These

lessons form the basis of the understanding of the role played by demonetization, response to the pandemic, and tightening cycles in market-based valuation.

Substantive influences are also made by structural changes in monetary regimes. (Bianchi et al., 2009) disclose that a moderate adjustment of long-term interest rates through the credibility of the policies also reduces uncertainty hence high valuation multiples may be sustained. Their results bring home the fact that the effectiveness of policy shock depends on the overall credibility of the institution, which becomes increasingly critical in terms of the pandemic and the tightening cycles afterwards. Furthermore, (Forbes et al., 2024) provide evidence on the international level that demonstrates that global rate cycles are transferred to emerging economies and risk-premium adjustments are produced, which are capable of creating sectoral adjustments in the value of domestic conditions even under unstable circumstances.

The valuation theory further puts one in perspective of the effects of macro shocks on firm level prices. (Damodaran, 2012) argues that multiple like P/E and EV/EBITDA incorporate expectations with regard to growth, profitability, risk, and the cost of capital; therefore, any factor that changes the discount rate, i.e., shifts in repo-rates or tightening liquidity, has a direct effect on the results of the valuation. This view is re-emphasized using the rational of interpreting P/E ratios as given by implied expected returns; whereby high discount rates reduce the ratio and the reverse, low discount rates inflate the ratio (Easton, 2004). (Schueler, 2020) continues to argue that multiples need to be examined in the context of their economic environment, with the writer observing that the disturbances with earnings stability, leverage trend, or the market environment make valuation assessments difficult to perform.

It is the credit-risk channel, which supports the theoretical basis. (JARROW & TURNBULL, 1995) segment their pricing strategy of credit sensitive securities that have sensitivity to macroeconomic or policy changes and show that the perceived likelihood of default and consequently pricing of asset subject to risk are affected. The paradigm is of particular relevance in the case of banking institutions, in the valuations of which are reflected the expectations about the quality of credit, provisioning, and balance sheet strength. It is also relevant in the case of manufacturer of automobile, which relies on consumer financing thereby at risk of credit availability via the channels of demand.

The comparative methodology is explained by sector-specific theoretical perspectives. (Coccorese, 2009) clarifies the influence of market power on banks pricing behaviour, and thus

the rate of transmission of the monetary policy into the lending spreads, as well as profitability. Theoretical knowledge provided by (Liu, 2019) and (Alexander & Al-Malkawi, 2023) in the automobile industry focuses on the factors of demand cycles, dependency on financing, and cost structure, therefore making automobiles valuations highly dependent on consumer behaviour as well as macroeconomic factors. These views suggest that the same monetary shocks would create asymmetric effects in the valuations between the two sectors.

Taken together, the theoretical literature suggests that valuation multiples are modulated in reaction to variations either in discount rates, earnings expectation, risk perception, and macroeconomic profits. Monetary shocks have a different effect on these parameters among industries depending on the structural differences. Therefore, the theoretical support to the comparative valuation analysis of banking and automobile industries in the major episode of shock in India is supported by asset-pricing models, transmission mechanism and valuation models.

2.11. Conceptual Framework/ Model Development

The conceptual framework embraced in this paper is namely based on the assumption that significant macroeconomic and monetary shocks affect the firm level valuation behaviour via the already established monetary transmission and asset-pricing processes. Each of these shocks, namely, demonetization in 2016, the COVID-19 pandemic in 2020, and the series of monetary tightening since 2022 have changed the liquidity situation, the interest rate expectations, the sectoral patterns of demand, or the way investors felt, and changed the valuation multiples in Indian industries. The empirical literature is very clear that the prices of assets are sensitive to the monetary shock, which shows the changes in discount rates, risk premia, and expected earnings (Alessi & Kerssenfischer, 2019) (Banerjee et al., 2018) (Behera et al., 2023). Consistent with this, monetary-policy theory emphasizes the role played by policy-guidelines, interest-rate changes, and liquidity operations on macro-financial stability (Clarida et al., 2000) (Rigobon & Sack, 2004). The present model is grounded on these theoretical and empirical grounds.

In this theory, economic and monetary shocks are discussed as main independent factors, which convey the impacts via four canonical channels, among which the interest-rate channel, liquidity channel, demand channel and expectations channel are considered. The interest-rate channel reflects the impacts of adjustments of the repo-rates on the costs of borrowing, discount

rate as well as the cost of capital to the companies, and hence, impacts on the valuations according to the findings of (Amrutkar et al., 2022) and the dividend-discount insights of (CAMPBELL & SHILLER, 1988). This liquidity channel reflects the impact of credit access and system-wide liquidity -related conditions on investor behaviour and corporate profitability, especially in the intervention periods, like in the cases of LTRO operations, relying on the studies of (Rigobon & Sack, 2004) and the evidence of demonetization offered by (Anoop et al., 2018). The demand channel also puts emphasis on changes in consumption trends and production cycles that are particularly relevant when it comes to automobile companies, in which works by (Alexander & Al-Malkawi, 2023) and (Klein & Machlica, 2021) indicate sector-specific sensitivities. Lastly, the expectations channel deals with the quality of market beliefs about future returns, future growth, and risks following theoretical teachings of (Fama & French, 1993) and emerging pandemic-oriented studies like (Mohanty & Mishra, 2024).

The given framework acknowledges the fact that transmission effects are not evenly distributed among all sectors. Sector-specific features are moderating factors affecting the extent of absorption and response of firms to macroeconomic shocks. Examples of these structures include the banking industry which is structurally adjusted to the monetary policy because it relies on the interest spread, the regulatory capital and liquidity control. (Coccorese, 2009), (Dayag & Trinidad, 2019), and (Marisetty & KM, 2025) establish that the reactions of banks valuations are closely related to any changes in interest rates and liquidity environments. Conversely, the automobile industry is more sensitive to the changes in demand, accessibility of consumer-finance, change in import costs and production cycles. As (Liu, 2019), (Keswani et al., 2024), and (Sarin, 2024) argue, the impact of economic shocks on the value of cars, like disruptions in the supply chains or a tightening of the credit is significant. These moderating features offer a theoretical reason as to why the banking and automobile industries are analysed separately.

The dependent variables of this model consist of the valuation multiples price-to-earnings (P/E) and enterprise value-per-EBITDA (EV/EBITDA) which are popular market expectations proxies, quality of earnings and risk perception. The studies conducted before underline the applicability of such measures to assess the expectations of discounted-cash-flow and peer-relative pricing (Damodaran, 2012) (Easton, 2004) (Schueler, 2020) (Gupta, 2018). The multiples of valuation used to reflect the current performance and the future is well positioned to internalise the impact of monetary and macroeconomic shocks on financial markets, and

thus, they are unique. The composite conceptual model, therefore, has macroeconomic shocks at the top, followed by four transmission mechanisms which can smooth out the fundamentals of the firm and investor perceptions. These relationships are also influenced by sector specific moderators, which allow drawing a comparative analysis of banking and automobile companies. The final performance of the model is uppercase through firm-level valuation performance, which is operationalised in P/E and EV/EBITDA ratios. Such architecture will provide a comprehensive, theory-driven basis on hypothesis development and empirical study between 2016 and 2024.

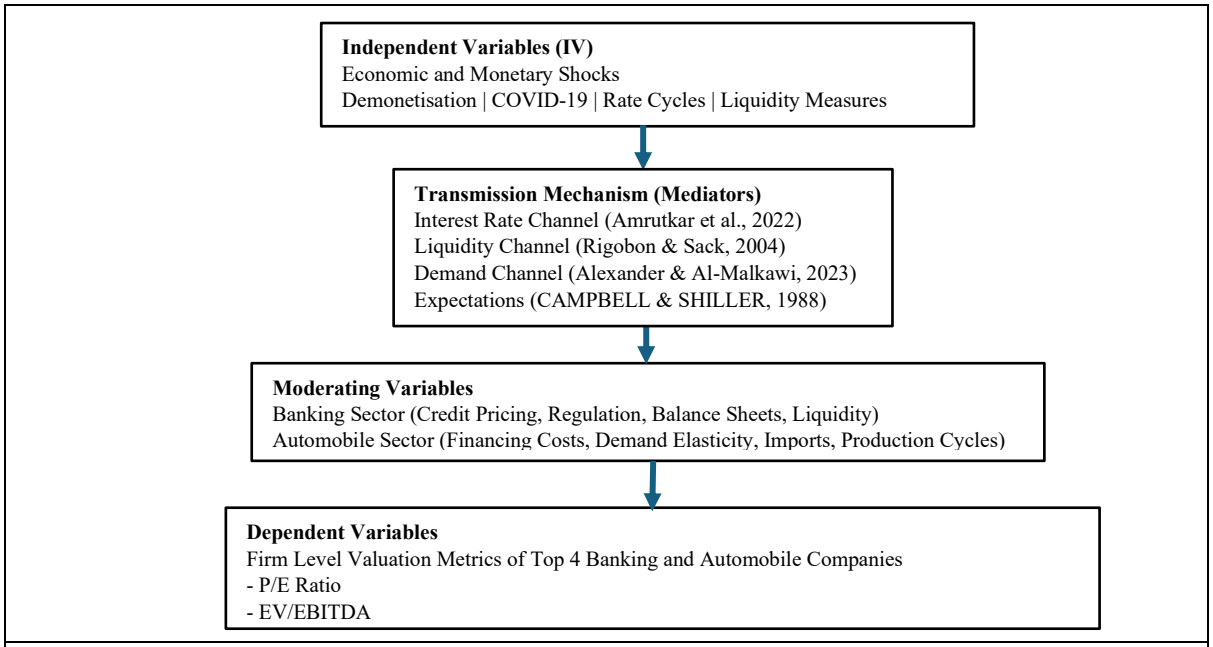

Figure 1: Proposed Conceptual Model *(Alessi & Kerssenfischer, 2019)*

2.12. Hypothesis Development

The hypothesis made in this analysis bring out the theoretical assumptions and practical evidence stated above. The literature review on monetary policies has always demonstrated that the changes in policy rates moderated the discount rates, capital costs and the expectation of investors; thus affects the market valuations. The findings of (CAMPBELL & SHILLER, 1988) together with (Gode & Ohlson, 2000) and (Alessi & Kerssenfischer, 2019) all prove that monetary tightening does not leave equity valuations unchanged, but changes them through adjustments in earnings and shifts in risk-premiums. Based on this, the initial hypothesis is as follows.

$H01_0$: Monetary policy tightening does not significantly affect firm valuation multiples (P/E and EV/EBITDA).

$H01_1$: Monetary policy tightening significantly affects firm valuation multiples (P/E and EV/EBITDA).

It is also pointed out in the existing literature that the banking industry is sensitive to monetary policy in that any change of interest rates is directly converted into a change in the net interest margins, the cost of funds and the liquidity environment. (Coccorese, 2009) and (Dayag & Trinidad, 2019), and (Marisetty & KM, 2025) stress that the banks are more concerned with the policy changes than non-financial sectors. These dynamics in the sector support the second hypothesis.

$H02_0$: The effect of monetary policy tightening on valuation multiples does not differ between banking and automobile firms.

$H02_1$: The effect of monetary policy tightening on valuation multiples differs between banking and automobile firms.

The 2016 demonetisation episode caused spare liquidity shifts, which impacted the operation of financial markets and behaviour within the sector. The third hypothesis is supported by empirical findings of (Anoop et al., 2018) and (Upadhyay & Suvarna, 2018) who provided evidence that the demonetisation brought about instant shifts in the way industries are valued.

$H03_0$: Demonetisation did not cause a significant change in firm valuation multiples.

$H03_1$: Demonetisation caused a significant change in firm valuation multiples.

Studies that were done in the period of demonetisation also reveal that banks had gained in terms of larger deposits and greater flow of formalisation, but cash intensive industries were to experience more severe operations pressures. This difference effect is in line with the results of (Anoop et al., 2018), which underlie the fourth hypothesis.

$H04_0$: The impact of demonetisation on valuation multiples does not differ between banking and automobile firms.

$H04_1$: The impact of demonetisation on valuation multiples is different for banking and automobile firms.

In 2020, the triggering of COVID-19 brought terrible effects to the economy, supply chains, and financial markets. Research like (Behera et al., 2023) and (Mohanty & Mishra, 2024) reports a high volatility and structural rupture of valuation behaviour in this time, thus, forming the basis of the fifth hypothesis.

$H05_0$: The COVID-19 period did not significantly affect firm valuation multiples.

$H05_1$: The COVID-19 period significantly affected firm valuation multiples.

The COVID research has shown that more significant demand shrinks, production interruptions and supply-side bottlenecks were felt in automobile companies than in financial institutions. (Klein & Machlica, 2021) and (Alexander & Al-Malkawi, 2023) share the increased vulnerability of the automobile industry at this time, proving the following hypothesis.

$H06_0$: The COVID-19 impact on valuation multiples does not differ between banking and automobile firms.

$H06_1$: The COVID-19 impact on valuation multiples differs between banking and automobile firms.

The liquidity measures, such as LTRO and specific easing interventions have been found to impact the behaviour of the stock-market, especially in emerging markets. Recent work by (Chundakkadan & Sasidharan, 2022) as well as (Rigobon & Sack, 2004) highlight the significance of liquidity shocks in the experience of valuation. This literature inspires the seventh hypothesis.

$H07_0$: Changes in the RBI repo rate do not significantly affect firm valuation multiples.

$H07_1$: Changes in the RBI repo rate significantly affect firm valuation multiples.

The business model of banks makes them more liquidity-shock responsive as they depend on the flow of funding and regulatory liquidity standards. The results obtained by (Coccorese, 2009) and (Marisetty & KM, 2025) support the presupposition that the funds of a bank are quite delicate to the liquidity situations shaping the eighth hypothesis.

$H08_0$: Repo rate changes do not have a stronger impact on banking firms than automobile firms.

$H08_1$: Repo rate changes have a stronger impact on banking firms than automobile firms.

Lastly, empirical research highlights that sectoral heterogeneity is at the centre of the long-term valuation behaviour. (Ghosh, 2019) and (Keswani et al., 2024) demonstrate that industries that are characterized by varied structural aspects have different valuation dynamics upon response to macroeconomic trends, thus the motivation behind the ultimate hypothesis.

$H09_0$: Banking and automobile firms do not exhibit significantly different valuation behaviour over the period 2016–2024.

$H09_1$: Banking and automobile firms exhibit significantly different valuation behaviour over 2016–2024.

The study also constructs clear cross-sector mean comparison hypotheses in addition to regression-based hypotheses to test the hypothesis that valuation multiples vary systematically between banking and automobile companies. In preceding literature, it is emphasized that industries that have unique financial structures, capital intensity, and demand sensitivity are heterogeneous in their valuation behaviour during macroeconomic and monetary regimes (Ghosh, 2019) (Alexander & Al-Malkawi, 2023) (Keswani et al., 2024). However, Indian research seldom engages in formal statistical tests to compare valuation multiple across market lines.

Based on that, $H10_0$ is formulated to examine the hypothesis that the means of P/E and EV/EBITDA ratios of banking and automobile companies change as per the study period. These theories are tested via a rate of two-sample t-test under unequal variances (Welch t -test), which is believed to be suitable since the sample sizes are not equal across industries, and the variances may not be equal.

$H10_0$: There is no significant difference in valuation multiples between banking and automobile firms (P/E and EV/EBITDA).

$H10_1$: There is a significant difference in valuation multiples between banking and automobile firms (P/E and EV/EBITDA).

This hypothesis is tested using a two-sample Welch t-test separately for P/E and EV/EBITDA ratios.

3. Chapter 3. Research Methodology

3.1. Research Design and Approach

The study will assume the quantitative and longitudinal research design to question how the role of significant macroeconomic and monetary perturbation has on the valuation multiples of Indian banking and automobile businesses in the years 2016-2024. The quantitative orientation is also supported by the fact that the need relies on numerical financial metric, such as price-to-earnings ratios, enterprise value to EBITDA multiples, interest rates, liquidity metrics, and macroeconomic variables, thus making the study to rest with the current asset-pricing and monetary transmission theories that predicts statistical evidence in the process of evaluating the consequences of various policy enactments on market valuation (CAMPBELL & SHILLER, 1988) (Fama & French, 1993) (Gode & Ohlson, 2000). The longitudinal structure allows one to detect changes in valuation over a long period of horizon, which is required to determine the disruptions caused by the demonetisation, the COVID-19 pandemic, and the tightening cycle of monetary policy. These stimulated different structural changes in the Indian economy as it was described in the empirical literature (Anoop et al., 2018), (Behera et al., 2023), and (Tomar & Kesharwani, 2022), thus making a time-series approach a prevailing consideration to consider how the sector responds.

The study has also assumed a comparative sectoral design because it has focussed on two sectors, which vary significantly in financial structure, policy sensitivity as well as demand and credit cycle exposure. The banking industry which lies at the heart of the monetary transmission is quite different with the automobile industry where consumer demand, availability of funds, production cycle and cost structure determines its dynamics. This opposition provides a solid theoretical found on to investigate asymmetric effects of transmissions that can be seen as stressed in previous literature (Ghosh, 2019) (Keswani et al., 2024). Placing the inquiry in this comparative context, the study measures the presence of a (structural) difference in how risk is recalibrated across macroeconomic regimes by industries that are structurally different.

The analysis also utilizes a sector-specific comparative statistical test using two-sample t-test with unequal variances (Welch t-test) to carefully evaluate the possible differences between the banking and automotive firms as regards to valuation behaviour. The process is used to supplement the regression model to provide an autonomous inferential confirmation to sectoral difference in valuation multiples.

3.2. Data Collection Methods

The methodology is limited to the secondary sources, as cross-referenced public repositories will provide the most reliable sources of macroeconomic variables, financial indicators, and valuation multiples. relying solely on the secondary sources is optimal to maintain objectivity, guarantee the replicability, and comply with the set of norms in the research of valuation and monetary policy. Corporate filings of the NSE and BSE are pried into firm-level metrics (P/ E ratios, EV/ EBITDA multiples, earnings, and financial performance) with the help of databases like Money Control, Screener, and CMIE Prowess. Such sources provide audited and standardized financial data that can fit the empirical basis of the valuation research (Damodaran, 2012) and (Schueler, 2020).

The reserve bank of India and the ministry of statistics and programme implementation provide macroeconomic and policy level information such as changes in the repo rates, liquidity interventions like LTRO operations as well as inflation and industrial output. The datasets found an extensive use in transmission and policy-shock studies that are being carried out in India (Amrutkar et al., 2022) (Chundakkadan & Sasidharan, 2022). The event specific information is collected in relation to demonetisation announcements, lockdowns introduced by COVID-19 and communications from policies which is documented in RBI press releases, Ministries of Finance circulars, peer reviewed studies (Anoop et al., 2018)(Upadhyay & Suvarna, 2018) (Mohanty & Mishra, 2024). Since the only quantitative indicators to be considered are those that can be found publicly, no primary data will be needed.

3.3. Sampling Technique

The current research uses purposive sampling; the firms will be sampled on the basis of representativeness of the particular sector and availability of consistent uniform data over the period of nine years between 2016 to 2024. This design structure is justified, since the analysis purpose needs big, continually reporting organizations that are reliable to capture valuation dynamics in the sector and supply uniform valuation multiples, including price-to-earnings (P/E) and enterprise value-to-EBITDA (EV/EBITDA).

The interest in how industries respond to macroeconomic perturbations in sector selection is motivated by the need to study those with different structural and behavioural characteristics. The existing body of literature highlights how the banking industry is closely connected to the monetary policy through such mechanisms as credit creation, liquidity provision based on the

mechanisms and the transmission of interest rates (Coccorese, 2009) (Dayag & Trinidad, 2019). On the other hand, the automobile industry is more vulnerable to demand factors, financing expenses, dependability on the supply chain, and customer mood, which is supported by (Alexander & Al-Malkawi, 2023) and (Klein & Machlica, 2021). This contrast provides a suitable background when it comes to analysing sectoral discriminations concerning valuation responses to macroeconomic and monetary shocks.

Selection of the firms was done by strict rules, which included continuous listing throughout the duration of the study, transparency of financial reporting and accessibility of valuation multiples to the public. Market capitalisation was used as a preliminary screening predictor; the capitalisation was higher, the more industry leading, the greater was the scale of its activities, the more covers by the analysts, the better was the market liquidity, and thus more predictable and economically meaningful measures of the valuation.

In line with it, the sample of the banking sector will include HDFC Bank, ICICI Bank, State Bank of India and Kotak Mahindra Bank, which have all maintained competitive positions as regarded to be on the forefront of the Indian market as regards to market capitalisation and operational magnitude. Institutional investors and analysts closely monitor these institutions, making the dynamics of their valuation reflect what is happening in the sector in response to the actions of the monetary policy.

Maruti Suzuki, Mahindra and Mahindra, Bajaj Auto, and Tata Motors were initially taken into account in the automobile industry due to their popularity and topicality on the market. Tata Motors was not included in the end empirical study since the valuation multiples of the company like P/E and EV/EBITDA could not be calculated in many years of the study period successfully due to negative and extremely fluctuating earnings. Since this study focuses specifically on valuation- multiple behaviour, it was observed that the entities in which the valuation- multiple ratios are not structurally-defined or economically informative were not recorded in order to protect the methodological validity. Therefore, the final sample of the automobile industry will consist of Maruti Suzuki, Mahindra and Mahindra, and Bajaj Auto, all of which will offer reliable and interpretable data of valuation during the research period.

The last sample containing four banking companies and three automobile companies provides the balance and analytically manageable structure to explore cross sector valuation behaviour,

yet has retained the internal consistency, data-validity, and such economic interpretability of valuation multiples.

The seven companies that were chosen to be added to this research were included in the NIFTY 50 index for a significant part of the period of observation, and they are the biggest, most liquid and systemically significant organizations in their respectable industries. Thanks to the inclusion they make the studied valuation multiples reflect market-wide expectations not idiosyncratic firm-specific anomalies, making them appropriate proxies of the valuation behaviour in the sector.

3.4. Data Analysis Methods

A systematic series of quantitative analytic strategies in this study is used to examine the impact of both monetary policy interventions and major macroeconomic shocks on firm-levels of valuations multiples. The analytical process is designed in a way that provides a robustness, interpretability and congruency with the research objectives.

The first step of the analysis is the descriptive statistics that is used to provide the summary of the central tendency, the dispersion, and the distributional nature of such salient variables as the Price-to-Earnings (P/E) ratio, the EV/ EBITDA ratio, and the policy repo rate between 2016 and 2024. The given preliminary step provides a preliminary insight into the valuation behaviour on a firm and industry level and highlights discrepancies in volatility and size between a banking and automated car business.

This is followed by a correlation analysis that is done to investigate the linear relationships between the valuation multiples, sector classification, and the macroeconomic variables. The correlation matrix is used as a diagnostic to specify the direction and strength of the associations as well as to test the possibility of the presence of multicollinearity before the estimation of regression. The fact that there was no high correlation between pairs gives reason to the suitability of the data to the multivariate regression analysis.

The basic empirical work is based on Ordinary Least Squares (OLS) regression, a linear estimation technique that minimises the sum of squared residuals between observed and predicted values. The reason why OLS is selected is that the dependent variables (P/E and EV/ EBITDA) are observable data at the firm level (directly). These variables reflect directly measure the valuation of firms and the independent variables are macroeconomic ones. Latent

constructs or unobservable theoretical aspects are not involved in the study and therefore complex structural modelling procedures like the SEM or the SPSS AMOS are unnecessary. Latent variables are hypothetical, immeasurable variables, e.g., investor sentiment or risk perception, and thus require a structural modelling method, e.g. Structural Equation Modelling (SEM). Conversely, all the variables used in the current study can be directly observed thus making such methods unnecessary. The OLS regression use coincides with empirical literature of valuation and monetary transmission analysing valuation multiples with available financial and macroeconomic data.

Discrete changes in macroeconomic regimes, including demonetisation, the COVID-19 pandemic, and a long phase of monetary tightening use dummy variables to represent them. The model allows the existence of the effects of structural shock without being subjected to linear continuity, thus consistent with event-based literature of monetary transmission.

There is a four regression models estimation. In models 1 and 2, the fixed effects of the monetary policy and macroeconomic shocks on P/E and EV/EBITDA are evaluated using pooled firm-level data as the models utilize a cross-section of equity-related variables. The next two models 3 and 4 are expansions of the baseline model incorporating interaction terms between sectoral classification and each of the key macroeconomic variables and thus allow the analysis to reflect the sector-specific disparity in valuation responses between banking and automobile companies.

The paper also uses a two-sample t-test of unequal variances (welch t -test) to augment the results of the regression to give an independent statistical test of industry disparities. This is a test that is conducted in order to compare the mean valuation multiples of the banking and automobile companies in relation to the period of study. The t-test used by Welch is more favourable than the traditional pooled t-test in that it does not presuppose the similarity of the variances among groups, which is why it is more appropriate to be used in the present case, considering the existing differences in the rate of valuation volatility in the two sectors. The results of this test provide strength to the result of the regression interaction since it provides direct inferential data regarding the differences in cross-sector valuation.

All statistical processes are performed with the use of Microsoft excel to clean and preliminarily analyse the data and SPSS to estimate econometric and conduct inferential tests. The combination of descriptive analysis, correlation testing, regression estimation, and cross-

sector mean comparison is a guarantee of a thorough and methodologically sound analysis of valuation behaviour in a variety of monetary and macro-economic regimes.

3.5. Summary

The chapter outlines a strict quantitative research strategy that is meant to examine the effect of significant macroeconomic shocks on valuation multiples within the Indian banking and automobile industries. The combination of the secondary data, the purposive sampling, market-capitalisation justification and systematic combination of the descriptive, event-based, regression and structural-break methods provide the study with the sufficient level of empirical and analytical thoroughness. Methodological approach is aligned to the theoretical framework as well as the hypotheses already developed in previous chapters, and thus offers a sound basis to the empirical analysis introduced in Chapter 4.

4. Chapter 4. Data Analytics and Findings

4.1. Overview of Data and Analytical Approach

This study draws on annual firm-level data between the years 2016 and 2024 to examine how the monetary policy and macroeconomic shocks affect the equity valuation multiples in the Indian market. The analytical concentration narrows down to the banking and automobile industries, as they have differentiated sensibilities to the interest rate movements, credit conditions as well as the dynamics of economic cycles.

The last sample is consisting of seven listed Indian companies. The banking group includes HDFC bank, ICICI bank, State Bank of India and Kotak Mahindra bank, whilst the automobile group entails Maruti Suzuki, Mahindra and Mahindra and Bajaj Auto. A sector dummy variable is created and it is used to represent 1 as banking enterprises and then 0 as automobile firms. The research has up to 63 firm-year observations. The Price-to-Earnings ratio (P/E) and EV/EBITDA are dependent variables since they were obtained when using Screener (*Screener*, n.d.). The information about policy repo rates is obtained in case of official publications of the Reserve Bank of India (*Reserve Bank of India*, n.d.). The repo rate variable reflects the annual policy rates set by the Reserve Bank of India and thus it shares similarity among firms in a certain year. Based on this, the number of observations of the repo rate is equal to the number of firm-year observations (N = 63), hence adhering to traditional panel data assumptions.

The Tata Motors was inhibited out of the final sample not because of incompleteness of data but because they had valuation infeasibility. The firm also recorded the negative or very volatile net income in a number of the years during the study period making the P/E ratio economically worthless. Also, negative or distorted EBITDA, high leverage, and high restructuring activity are those periods which make the EV/EBITDA interpretation problematic. The related literature on valuation cautions the use of multiples in that case because they do not show the underlying economic value. This led to the exclusion of Tata Motors as a way of maintaining the similarity and comparable values among companies as guided by the principles of valuing companies (Damodaran, 2012).

The P/E ratio of State Bank of India has two observed firm-years 2018 and 2019 that are unavailable due to accounting distortions during that period that were caused by the restructuring. Some of the firm-year observations, namely the ones related to the State Bank of India in 2018 and 2019, exhibit either non-computed or economically non-significant price-to-

earnings ratios due to high earnings volatility or low profitability. Although these observations are not removed from the dataset, SPSS conveniently omits them in analyses, where the price-to-earnings ratio is being used as the dependent variable, correspondingly giving smaller effective sample size to such models. This methodology approach prevents data corruption and prevents artificially altered data. Therefore, the regressions based on P/E as the dependent variable are estimated based on 61 observations whilst those based on EV/EBITDA apply the complete 63 observations. This difference in sample size among models is standard in empirical research of firms and has no bias on the empirical results.

Binary dummy variables are used to capture macroeconomic and policy shocks. A dummy of demonetisation will be 1 in the year 2016 and 0 during the rest of the years. COVID -19 is a dummy of 2020 and 2021 indicating the economic assumption of the pandemic. To reflect the prolonged tightening that has been occurring after monetary environmentally-friendly mass measures after pandemic, the higher level of 2022 to 2024 is treated as a monetary tightening dummy. These variables denote different policy regimes and structural shocks what concerns the period of study. The use of dummy variables enables the model to capture the discrete changes of regime not assuming continuous linear responsiveness, a method that is consistent with the existing literature of event-based monetary transmission and valuation.

Preliminary analysis and data cleaning was performed using the Microsoft Excel whereas all econometric estimations were performed using SPSS. To develop the estimates, Ordinary Least Squares regression was used to estimate four models. There was no use of structural equation modelling software like SPSS AMOS since all the variables of the research are directly observable, and the study design does not imply the presence of any latent variables or mediating factors. This analysis method is therefore relevant and in line with the aims of the research.

4.2. Descriptive Statistics

In this section, the most important variables of the given research were supplied with descriptive statistics to represent the picture of distribution, central tendency and dispersion of the valuation multiples and policy repo rate during the period of the sample.

Variable	P/E Ratio	EV/EBITDA	Repo Rate %
N	61	63	63
Mean	25.29	17.88	5.60
Median	22.40	18.30	6.00
Std. Deviation	11.15	4.95	1.08
Variance	124.42	24.54	1.16
Minimum	9.60	6.20	4.00
Maximum	69.00	32.40	6.75

Table 1: Descriptive Statistics of Pooled Samples for Key Variables generated using Excel

Note: State Bank of India has 2018 and 2019 all-time low P/E ratios because of accounting distortions caused by restructuring. Therefore, P/E based statistics are calculated by the pooled sample of 61 and 34 observations in the banking sector subsample. Although the repo rate is the same among firms in the same year, it is also reported with N=63 to maintain the same consistency with the firm-year observations of the panel dataset.

The descriptive statistics of the pooled sample consisting of the banking firms and the automobile firms are listed on Table 1. P/E ratio is present in 61 observations and EV/EBITDA and repo rate present in 63 observations. The pooled sample demonstrates a mean P/E ratio of 25.29 and a median of 22.40 which shows that it is skewed to the right because of significant episodes of high valuation of individual firm-year observations. The value of the standard deviation 11.15 and variance 124.42 are large indicators of a high dispersion in the earnings-based valuation within the study period. Minimum P/E ratio is at 9.60 whilst maximum is at 69.00 thus depicting valuation compression and expansion periods.

The average EV/EBITDA multiple of the consolidated sample equals 17.88, and the median of the sample equals 18.30, which implies the relatively symmetrical distribution unlike that of P/E ratios. The variance is 24.54 and standard deviation is 4.95 and means that there is less dispersion of the business valuation based on the enterprise value that is related to earnings volatility. The range of EV/EBITDA values is between 6.20 and 32.40. The median and mean repo rate in the study period is 6.00 and 5.60 respectively, which represents both the period of accommodative and tightening monetary policy. The range of the repo rate is 4.00 -6.75 and the standard deviation stands at 1.08.

Variable	P/E Ratio	EV/EBITDA	Repo Rate %
N	34	36	36
Mean	24.21	19.84	5.60
Median	22.55	19.40	6.00
Std. Deviation	9.97	2.67	1.08
Variance	99.42	7.10	1.17
Minimum	9.60	13.90	4.00
Maximum	47.90	26.20	6.75

Table 2: Descriptive Statistics for Banking Sector generated using Excel

Note: State Bank of India has 2018 and 2019 all-time low P/E ratios because of accounting distortions caused by restructuring. Therefore, P/E based statistics are calculated by the pooled sample of 61 and 34 observations in the banking sector subsample.

Table 2 is a descriptive-statistics of the banking industry, which includes 34 P/E observations and 36 EV/EBITDA observations. The average P/E ratio of banking companies is 24.21 with the median and 22.55. The values of the standard deviation (9.97) and the variance (99.42) demonstrate the existence of moderate dispersion in the bank valuations compared to the pooled sample. The values of P/E ratios in the banking sector vary between 9.60 and 47.90 to indicate variations in profitability and quality of assets and growth prospects between firms and years.

On average, the banking firms have a multiple of 19.84 on the ratio of EV to EBITDA with a median of 19.40. Dispersion level is fair with a standard deviation of 2.67 and a variance of 7.10 which meant that there was more value stability at the enterprise level in the banking sector. The EV/EBITDA analysis is between 13.90 and 26.20. The banking subsample repo-rates statistics are the same because of the homogeneous implementation of monetary policy across industries, with a mean value of 5.60, median of 6.00, and a value difference of 4.00 to 6.75.

Variable	P/E Ratio	EV/EBITDA	Repo Rate %
N	27	27	27
Mean	26.64	15.27	5.60
Median	22.40	13.80	6.00
Std. Deviation	12.55	6.05	1.09
Variance	157.52	36.61	1.18
Minimum	10.26	6.20	4.00
Maximum	69.00	32.40	6.75

Table 3: Descriptive Statistics for Automobile Sector generated using Excel

Table 3 presents the descriptive statistics of the automobile industry which comprise of 27 observations of P/E and EV/EBITDA. Mean P/E of automobile companies stands at 26.64 which is higher than the mean of banking industry with the median of 22.40. A standard deviation value of 12.55 and variable of 157.52 implies that the automobile values are more dispersed hence it is more price sensitive to cyclical changes in demand and price of input costs and the entire macroeconomic environment. The automobile industry has P/E ratios between 10.26 and 69.00 with a strong solid valuation volatility in the study period.

The median and average EV/EBITDA multiple of automobile companies is 13.80 and 15.27 respectively which is lower than the banking industry figures. Compared to banking, it has a greater level of dispersion shown by a standard deviation of 6.05 and a variance of 36.61. The range of EV/EBITDA values is 6.20 to 32.40, so there is a great difference in the valuations of the enterprises during the time of the years and firms. Statistical results of repo rate in the automobile industry are similar to that of the sample, with the mean of 5.60, median 6.00, and a range of 4.00 to 6.75.

Key Interpretation:

The descriptive statistics show that distinct differences in valuation behaviour among sectors exist. The P/E and EV/EBITDA ratios of the automobile companies are more volatile whereas the enterprise-value based valuations at the bank companies are relatively stable. Such trends present background indications of sector heterogeneity formalized in the later analysis through regressions.

4.3. Correlation Analysis

In this section, the Pearson correlation coefficients between the important variables employed in the study are given. The correlation analysis provides initial information on the direction and

strength of linear relationships and provides a diagnostic checkpoint before regression estimation on the possibility of multicollinearity.

	Sector_Dummy	P/E	EV/EBITDA	Repo_Rate
Sector_Dummy	1			
P/E	-0.108861802	1		
EV/EBITDA	0.459367648	0.574216761	1	
Repo_Rate	2.67013E-17	-0.030741273	-0.179994256	1

Table 4: Correlation Matrix of Key Variables generated using Excel

The correlation matrix (Table 4) shows that the relationship between P/E ratio and EV/EBITDA has moderate positive correlation of 0.574, meaning that the two valuation ratios are correlated but reflect different elements with regard to the valuation of firms. This substantially allows both measures to be used as independent variables in later regression analysis as a dependent variable.

The sector dummy, with the value of 1 representing the banking firms and 0 the automobile firms, has weak negative correlation of -0.109 with P/E. This denotes the fact that the average banking companies are expected to experience a bit low P/E ratios compared to the automobile firms during the period of study. Conversely, the sector dummy is averagely positively correlated with EV/EBITDA at 0.459, meaning that the banking companies tend to attract a premium based on enterprise-value on the average than the automobile companies.

The correlation of repo rate with P/E is so weakly negative at -0.031 which implies that there is zero linear contemporary relationship between movements on the policy rates and earnings-based valuation multiples at the aggregate level. The aspect of the repo rate and EV/ EBITDA is by -0.180 which showed a weak negative relationship amid interest rates and value based on enterprise and financing costs, represent a true position of interest rates.

Correlation of the sector dummy with the repo rate is also making sure that the variation in policy repo rate has zero correlation with selections to the sector and is equally applicable across the industries (2.67×10^{-17}). Notably, no correlation coefficient in the articles is greater than generally accepted levels related to the issue of multicollinearity. This implies that the explanatory variables can be contained in the regression models simultaneously without swelling the standard errors or estimating the coefficients.

The obtained outcome of correlation shows that there are no strong linear relationships between variables. This proves the appropriateness of the dataset toward multivariate regression analysis. In the next section, we have provided the baseline regression models which are used to study the effect of monetary policy and the macroeconomic shock on valuation multiples.

4.4. Regression Results – Baseline Models

The overall econometric specification to be applied in estimating the effects of the monetary policy, macroeconomic shock on valuation multiples is mentioned below before placing the baseline regression outputs.

$$Y(it) = \beta 0 + \beta 1 RepoRate(t) + \beta 2 Demonetization(t) + \beta 3 COVID19t + \beta 4 MonetaryTightening(t) + \varepsilon(it)$$

Y(it) denotes the valuation multiple of firm (i) in year t. In Model 1, Y(it) is measured as the Price-to-Earnings (P/E) ratio, while in Model 2, it is measured as EV/EBITDA. RepoRate represents the policy repo rate, while the dummy variables capture the demonetization period, the COVID-19 shock, and the monetary tightening phase. The error term ε(it) captures unexplained variation in valuation multiples.

A. Model 1: Impact of Monetary Policy and Macroeconomic Shocks on P/E Ratio

Model 1 studies how individual variables of monetary policy and macroeconomic shocks of great magnitude affect the Price-to-Earnings (P/E) ratio that is pooled in a combination of both banking and automobile companies. Ordinary Least Squares regression is used to estimate the model as the dependent variable is the P/E ratio, the explanatory variables are the repo rate, the demonetization dummy, and COVID-19 dummy, and the monetary tightening dummy.

	Variable	Unstandardized Coefficient (B)	Standard Error	t-value	p-value	
	Constant	47.18	12.415	3.8	<0.001	
	Repo Rate	–2.879	1.996	–1.442	0.155	
	Demonetisation Dummy	–4.448	5.009	–0.888	0.378	
	COVID-19 Dummy	–11.935	5.393	–2.213	0.031	
	Monetary Tightening Dummy	–7.434	3.564	-2.086	0.042	

Dependent Variable: P/E Ratios

Model 1 Summary:

Statistic	Value
R	0.322
R^2	0.103
Adjusted R^2	0.039
F-statistic	1.616
Sig. (F)	0.183
Durbin–Watson	0.955
N	61

Table 5: Regression Results of Model 1 generated using SPSS Statistics

Note: See that regression of P/E has two firm-years observation of State Bank of India (2018-2019) that are not used to obtain P/E value.

Table 5 gives the estimated coefficients of regression of Model 1. The model presents an R value = 0.322, R 2 = 0.103, adjusted R 2 = 0.039, which means that the macroeconomic variables factored accounted 10.3% of the fluctuation in P/E ratios. The F-ratio stands at 1.616 and the significance of 0.183, which indicates that the model is not jointly significant at regular levels. The observed value of Durbin-Watson, 0.955, does not suggest serious autocorrelation issues, which is reasonable, taking into account that annual firm-level data was used. The regression is estimated based on 61 observations, which are a product of two omitted firm-year P/E observations of State Bank of India.

The coefficient of the repository rate is estimated to be -2.879 with p -value=0.155 and this implies a negative relationship between the policy-rate changes and the P/E ratios but this relationship is not significant. The demonetization dummy has a coefficient of -4.448 at a p-value of 0.378 which is meaningless because it does not indicate statistically relevant long-term effect of demonetization on earnings-based valuations.

In comparison, the COVID -19 dummy shows a negative coefficient of -11.935 and is statistically significant at the 5 per cent level with a p-value of 0.031 implying a significant plummeting of the P/E ratios owing to the pandemic. Also obtained is the statistically significant negative influence of the monetary tightening dummy estimated using the

coefficient of -7.434 and a p-value of 0.042, indicating that a stable restrictive regime of the monetary policy compresses the earnings-based valuation multiples.

On the whole, according to Model 1, large macroeconomic shocks and extended periods of tightening show a statistically significant negative impact on P/E ratios, and the routine monetary policy rates changes cannot have a significant individual impact.

B. Model 2: Impact of Monetary Policy and Macroeconomic Shocks on EV/EBITDA Ratio

Model 2 explores the relationships between the variables of monetary policy and significant macroeconomic shocks and the EV/EBITDA, which is a valuation multiple based on enterprise value. This model is a reflection of the specification applied in Model 1; instead of P/E ratio as the dependent variable, the dependent variable is EV/EBITDA. Ordinary Least Squares regression with banking and automobile firm samples are used to estimate the model.

Variable	Unstandardized Coefficient (B)	Standard Error	t-value	p-value
Constant	27.817	5.482	5.075	<.001
Repo Rate	–1.648	0.884	–1.864	0.067
Demonetisation Dummy	–1.306	2.195	–0.595	0.554
COVID-19 Dummy	–3.487	2.359	–1.478	0.145
Monetary Tightening Dummy	0.642	1.54	0.417	0.678

Dependent Variable: EV/EBITDA Ratios

Model 2 Summary:

Statistic	Value
R	0.325
R Square	0.106
Adjusted R Square	0.044
F-statistic	1.711
Sig. F	0.16
Durbin–Watson	0.744
N	63

Table 6: Regression Results of Model 2 generated using SPSS Statistics

The values of the estimated regression of Model 2 are provided in Table 6. The model results in the determination of the R value of 0.325, the R 2 of 0.106, and the adjusted R 2 of 0.044, meaning that the macroeconomic variables included in the model account for 10.6-percent of the change in EV/EBITDA. F-statistics equal 1.711 corresponding to a level of 0.160, which shows that the model is not significant together at traditional levels. The value of Durbin-Watson is 0.744, which does not indicate any serious autocorrelation issues. The estimation of the regression has been carried out based on 63 observations, which represent the presence of the EV/EBITDA data on all the firms and years.

The coefficient of the repo rate is estimated at -1.648 with p-value of 0.067, which is also negative that the estimated coefficient is negative but does not reach statistical significance at conventional levels. This implies that enterprise-value based valuations are somewhat sensitive towards the modifications of the policy rate probably by the cost of capital and discounting of the cash flows in the future.

The coefficient of the demonetization dummy is -1.306 with the p-value of 0.554, and the effect of demonetization is not statistically significant in the long-term on EV/ EBITDA. Likewise, the coefficient of the COVID 19 dummy is negative (-3.487) with a p-value of 0.145 which means that the pandemic has no statistically significant impact on the valuation of enterprises based on enterprise-value after accounting for more macroeconomic variables.

The coefficient value of monetary tightening dummy is at 0.642 and the p-value of the coefficient is at 0.678; therefore, there is no statistically significant relationship between the long tightening periods and EV/EBITDA. In general, the findings of the Model 2 enable to conclude that EV/EBITDA is comparatively less affected by macroeconomic shocks than the earnings-based valuation multiples and slightly reacts to the difference in the policy repo rate.

4.5. Results of the Regression Interaction Models

Whereas the baseline models study the average effect of monetary policy and macroeconomic shocks on the valuation multiples, they implicitly presume that the overall effects are consistent across sectors. Since both banking and automobile companies have distinct structures, interaction models are approximated to study the assumption of variations in the impact of macroeconomic variables by sectors. Sectoral asymmetry is studied in this section by adding interaction terms between sectoral dummy and significant macroeconomic variables.

The overall econometric specification is specified below before the representation of issues of the interaction model.

$Y(it) = \beta 0 + \beta 1 RepoRate(t) + \beta 2 Demonetization(t) + \beta 3 COVID19(t) + \beta 4 MonetaryTightening(t) + \beta 5(RepoRate(t) \times Sector\ (i)) + \beta 6(COVID19(t) \times Sector\ (i)) + \beta 7(MonetaryTightening(t) \times Sector(i)) + \beta 8(Demonetization(t) \times Sector(i)) + \varepsilon(it)$

Y(it) represents the valuation multiple of firm i in year t. In Model 3, Y(it) is measured as the Price-to-Earnings (P/E) ratio. In Model 4, Y(it) is measured as the EV/EBITDA ratio. Sector is a dummy variable equal to 1 for banking firms and 0 for automobile firms. The interaction terms capture the incremental effect of macroeconomic variables on banking firms relative to automobile firms. The error term ε(it) represents unexplained variation.

A. Model 3: Sectoral Impact of Monetary Policy & Macroeconomic Shocks on P/E Ratio

Model 3 investigates the hypothesis that the influence of monetary policy and the macroeconomic shocks on the P/E ratios varies between a banking and an automobile firm. Ordinary Least Squares regression and the use of the interaction terms between the sector dummy and key macroeconomic variables are used to estimate the model.

Variable	Unstandardized Coefficient (B)	Standard Error	t-value	p-value
Constant	47.547	11.992	3.965	<0.001
Repo Rate	–3.466	1.981	–1.750	0.086
Demonetisation Dummy	3.115	7.245	0.43	0.669
COVID-19 Dummy	–11.213	6.358	–1.764	0.084
Monetary Tightening Dummy	1.761	4.925	0.357	0.722
Repo × Sector	1	0.78	1.282	0.206
COVID × Sector	–1.789	6.576	–0.272	0.787
Monetary Tightening × Sector	–16.575	6.42	–2.582	0.013
Demonetisation × Sector	–13.692	9.63	–1.422	0.161

Dependent Variable: P/E Ratio (With Sector Interaction Terms)

Model 3 Summary:

Statistic	Value
R	0.473
R^2	0.224
Adjusted R^2	0.104
F-statistic	1.873
Sig. (F)	0.084
Durbin–Watson	0.946
N	61

Table 7: Regression Results of Model 3 generated using SPSS Statistics

Note: P/E regressions are used without dropout firms, such as an unavailable P/E value in this case of State Bank of India (2018-2019).

Table 7 is a report of the regression of Model 3. The model provided gives an R value of 0.473, a R 2 of 0.224 and adjusting R 2 of 0.104, which means that the variation in P/E ratios can be attributed to the interactions between the sectors to 22.4 per cent. The statistically significant value of the F-test is 1.873, and the significance of the value is 0.084, indicating that the model has a significant margin of significance at the 10 -percent level which means the medium strength of the model in explaining when the interaction of the sectors is included. The Durbin-Watson of 0.946 which represents no serious issues of autocorrelation are obtained. There are 61 observations that are used to estimate the regression.

The coefficient on the repo rate equals -3.466 and p-value equals 0.086 thus showing a negative but statistically insignificant effect with P/E ratios for automobile companies in the baseline group. This implies that the valuations of automobile-sector are delicate to alterations in borrowing rates and demand states.

The coefficient of the COVID -19 a dummy is negative and is -11.213, which has a p-value of 0.084, showing sort of marginally significant adverse effect of the pandemic on the P/E ratios in the automobile-sector. However, the opposite does not hold true in the other case as the monetary tightening and demonetization dummy do not have a statistically significant direct effect on P/E ratios under the basic category.

One of the terms of interaction is significantly statistically significant and this is the Monetary Tightening x Sector interaction, with a coefficient of -16.575 and p -value of 0.013. This implies that the fall in P/E ratios of the banking companies was much more during the times of

prolonged tightening of the monetary policy compared to the case of automobile companies. Other phrases of interaction such as Repo x Sector (p = 0.206) and COVID x Sector (p = 0.787) and Demonetization x Sector (p = 0.161) are statistically insignificant and indicate that the most significant sectoral differences can be observed during long periods of tightening and not during short shocks.

On the whole, Model 3 findings suggest that adding some sectoral interaction effect substantially improves the explanatory power compared to a baseline P/E model and also expressed that meaningfully asymmetric effects of monetary tightening on valuation multiples across sectors.

B. Model 4: Sectoral Impact of Monetary Policy & Macroeconomic Shocks on EV/EBITDA Ratio

Model 4 builds on the interaction structure to EV/EBITDA to test the hypothesis that the effects of monetary policy and macroeconomic shocks vary among banking and automobile companies in the cases where valuation is specified through an enterprise-value based multiple. The model includes the terms of interaction between the sector dummy and the main macroeconomic variables and is estimated by Ordinary Least Squares regression.

Variable	Unstandardized Coefficient (B)	Standard Error	t-value	p-value
Constant	27.817	4.884	5.696	<0.001
Repo Rate	–2.106	0.807	–2.610	0.012
Demonetisation Dummy	0.732	2.962	0.247	0.806
COVID-19 Dummy	–5.622	2.595	–2.166	0.035
Monetary Tightening Dummy	1.462	2.013	0.726	0.471
Repo × Sector	0.801	0.307	2.61	0.012
COVID × Sector	3.735	2.664	1.402	0.167
Monetary Tightening × Sector	–1.435	2.578	–0.556	0.58
Demonetisation × Sector	–3.567	3.894	–0.916	0.364

Dependent Variable: EV/EBITDA Ratio (With Sector Interaction Terms)

Model 4 Summary:

Statistic	Value
R	0.582
R^2	0.339
Adjusted R^2	0.241
F-statistic	3.461
Sig. (F)	0.003
Durbin–Watson	1.081
N	63

Table 8: Regression Results of Model 4 generated using SPSS Statistics

Note: The sector dummy = 1 in banking firms and = 0 in automobile firms. Interaction terms win the incremental effects to banking firms as opposed to automobiles firm.

Table 8 is the Results of the regression using Model 4. The model gives an R value of 0.582, R 2 of 0.339, and an adjusted R 2 of 0.241, implying that 33.9 per cent. of the changes in EV/EBITDA is credited to the interplay of sectoral interaction. The F value is 3.461 and the p-value is 0.003 which confirms that the model is statistically significant under the 1 percent level. The value of Durbin-Watson is 1.081 to suggest that there are no serious issues with autocorrelations. The estimation of the regression is an estimate of 63 observations.

The repo rate coefficient, –2.106, has a p-value = 0.012 showing that it is statistically significant in the baseline category, indicating that EV/EBITDA valuations of automobile firms are sensitive to interest rate changes. However, the significant Repo Rate × Sector interaction reveals that this sensitivity is significantly weaker for banking firms.

The COVID-19 dummy has a coefficient of -5.622 and p-value of 0.035 which means that the pandemic has a statistically significant negative effect on EV/EBITDA of automobile companies. On the contrary, the direct impact of the demonetisation dummy (p = 0.806) and the monetary tightening dummy (p = 0.471) do not display statistically significant direct effects on EV/ EBITDA under the baseline classification.

The interaction terms that are statistically significant are the Repo Rate x sector interaction whose coefficient is 0.801 with a p -value of 0.012. This shows that the adverse interest-rate movements impact on EV/EBITDA is quite low with respect to banking firms as compared to automobile firms. The other terms of interaction, such as COVID x Sector (p = 0.167),

Monetary Tightening x Sector (p = 0.580), and Demonetisation x Sector (p = 0.364) are not significant. This implies that the sectoral variations in EV/EBITDA are mostly predetermined by dissimilar sensitivity to variations in interest rates, but not to particular macroeconomic shocks.

All in all, the findings of Model 4 show that the addition of sectoral interaction effects significantly enhances the ability of the model to explain compared to the initial EV/EBITDA model and creates a noticeable asymmetry in the responses of the interest-rate changes across the enterprise-value-based valuations across sectors.

4.6. Cross-Sectoral Comparison of Valuation Multiples: Welch's t-test Results

To supplement the regression-based analysis as well as to have an independent inferential determination of the sectoral valuation differences, this study uses a two sample Welch's t test to compare the mean valuation multiples of the banking and automobile companies. Welch t-test is the better choice than the traditional pooled t-test as it does not assume similar variance difference among groups and can withstand unequal sample sizes that are evident in present dataset.

P/E ratios and EV/EBITDA multiples are tested independently, with banking companies coded Sector=1 and automobile companies coded Sector=0.

A. Welch's t-Test Results of P/E Ratios

The banking sector consisted of 34 observations with an average price-to-earnings ratio of 24.21 for price-to-earnings, and the automotive sector included 27 observations with an average of 26.64. Although the mean of the automotive cohort is slightly higher, the t -test of the Welch shows that this observation is not statistically significant.

The Welch test statistic is -0.819 occurring against 48.914 degrees of freedom with a p-value of 0.417 two tailed. Since the p-value is greater than the traditional 5% level of significance, equal mean of the P/E ratios by sector cannot be rejected.

Moreover, a small and economically insignificant difference between the two sectors in terms of earnings-based valuation is demonstrated by the estimated effect size (Cohen d = -0.217).

Group Statistics

	Sector_Dummy	N	Mean	Std. Deviation	Std. Error Mean
P/E	1	34	24.214705882	9.9711441960	1.7100371219
	0	27	26.639259259	12.550626686	2.4153692318

Independent Samples Test

		t-test for Equality of Means				
				Significance		
		t	df	One-Sided p	Two-Sided p	Mean Difference
P/E	Equal variances assumed	-.841	59	.202	.404	-2.424553377
	Equal variances not assumed	-.819	48.914	.208	.417	-2.424553377

Independent Samples Test

		t-test for Equality of Means		
			95% Confidence Interval of the Difference	
		Std. Error Difference	Lower	Upper
P/E	Equal variances assumed	2.8823151721	-8.192052714	3.3429459606
	Equal variances not assumed	2.9594316150	-8.372017696	3.5229109417

Figure 2: Part 1 - SPSS Output of Welch's t-Test using P/E Ratio

Independent Samples Effect Sizes

				95% Confidence Interval	
		Standardizer[a]	Point Estimate	Lower	Upper
P/E	Cohen's d	11.181443944	-.217	-.723	.291
	Hedges' correction	11.326135844	-.214	-.713	.287
	Glass's delta	12.550626686	-.193	-.699	.317

a. The denominator used in estimating the effect sizes.
Cohen's d uses the pooled standard deviation.
Hedges' correction uses the pooled standard deviation, plus a correction factor.
Glass's delta uses the sample standard deviation of the control (i.e., the second) ...

Figure 3: Part 2 - SPSS Output of Welch's t-Test using P/E Ratio

Results:

H10₀ (P/E Ratio): Failed to reject as the p-value is 0.417 which is greater than 0.05.

There is no statistically significant difference in P/E ratios between the companies which are banking and automobiles in the years 2016-2024 as shown in Figures 2 and 3.

B. Welch's t-Test Results of EV/EBITDA Ratios

The t-test of EV/EBITDA of the Welch, on the contrary, indicates the statistically significant cross-sector difference. The mean EV/EBITDA multiple of banking firms is considerably greater than that of automobile firms, which also shows differences in the stability of capital structures, predictability of earnings, and enterprise-level valuation processes.

Welch statistic EV/EBITDA is equal to 3.660, the 33.589 degree of freedom, and the p-value of 2-tailed less than 0.001. The above outcome diverts decisively the null hypothesis of equals in the 1 per cent level.

The size of the test statistic and the very small p -value show that the difference in the enterprise-value based valuation between the banking and the automobile companies is significant statistically and economically.

T-Test

Group Statistics

	Sector_Dummy	N	Mean	Std. Deviation	Std. Error Mean
EV/EBITDA	1	36	19.836111111	2.6652421791	.44420702984
	0	27	15.274074074	6.0509117299	1.1644985054

Independent Samples Test

		t-test for Equality of Means			
				Significance	
		t	df	One-Sided p	Two-Sided p
EV/EBITDA	Equal variances assumed	4.039	61	<.001	<.001
	Equal variances not assumed	3.660	33.589	<.001	<.001

Independent Samples Test

		t-test for Equality of Means		
				95% Confidence Interval of the ...
		Mean Difference	Std. Error Difference	Lower
EV/EBITDA	Equal variances assumed	4.5620370370	1.1294495997	2.3035629794
	Equal variances not assumed	4.5620370370	1.2463453191	2.0280157379

Independent Samples Test

		t-test for Equality ..
		95% Confidence Interval of the ...
		Upper
EV/EBITDA	Equal variances assumed	6.8205110947
	Equal variances not assumed	7.0960583361

Figure 4: Part 1 - SPSS Output of Welch's t-Test using EV/EBITDA Ratio

Independent Samples Effect Sizes

		Standardizer[a]	Point Estimate	95% Confidence Interval Lower	Upper
EV/EBITDA	Cohen's d	4.4363899580	1.028	.493	1.556
	Hedges' correction	4.4918834841	1.016	.487	1.537
	Glass's delta	6.0509117299	.754	.209	1.287

a. The denominator used in estimating the effect sizes.
Cohen's d uses the pooled standard deviation.
Hedges' correction uses the pooled standard deviation, plus a correction factor.
Glass's delta uses the sample standard deviation of the control (i.e., the second) group.

Figure 5: Part 2 - SPSS Output of Welch's t-Test using EV/EBITDA Ratio

Results:

H10₀ (EV/EBITDA): Rejected as the p-value is lesser than 0.01 which is lesser than 0.05. There is a significant statistical variation in the EV/EBITDA multiples between the banking and automobile sectors within the research period as shown in figures 4 and 5.

C. Interpretation and Consistency with Regression Results

The t -test results of the Welch support the regression-based results of Models t -test 3 and 4. Although the difference in earnings-based valuation (P/E) results obtained is not statistically significantly different across sectors, the valuation based on enterprise-value (EV/EBITDA) reflects strong sectoral dispersion. Such trend means that capital structure, stability in fundings and enterprise level cash-flow expectations distinguish more between banks and automobile firms rather than earnings-based measures that are more vulnerable to current profit fluctuations. Welch test therefore offers independent support of sectoral heterogeneity of valuation behaviour, especially the valuation assessed at the enterprise level.

4.7. Summary of Hypothesis Testing Results

Table 9 summarises the results of the null hypothesis testing across the four regression models.

Hypothesis	Null Hypothesis	Model Tested	Decision
$H01_0$	Monetary policy tightening does not significantly affect firm valuation multiples (P/E and EV/EBITDA).	Tested in Model 1 & 2 (Tables 5 and 6)	$H01_0$ Rejected for P/E.

			$H01_0$ Failed to Reject for EV/EBITDA.
$H02_0$	The effect of monetary policy tightening on valuation multiples does not differ between banking and automobile firms.	Tested in Model 3 (Table 7)	$H02_0$ Rejected.
$H03_0$	Demonetisation did not significantly affect firm valuation multiples.	Tested in Model 1 & 2 (Tables 5 and 6)	$H03_0$ Failed to Reject.
$H04_0$	The impact of demonetisation on valuation multiples does not differ between banking and automobile firms.	Tested in Model 3 (Table 7)	$H04_0$ Failed to Reject.
$H05_0$	The COVID-19 period did not significantly affect firm valuation multiples.	Tested in Model 1 & 2 (Tables 5 and 6)	$H05_0$ Rejected for P/E. $H05_0$ Failed to Reject for EV/EBITDA.
$H06_0$	The COVID-19 impact on valuation multiples does not differ between banking and automobile firms.	Tested in Model 3 (Table 7)	$H06_0$ Failed to Reject.
$H07_0$	Changes in the RBI repo rate do not significantly affect firm valuation multiples.	Tested in Model 1 & 2 (Tables 5 and 6)	$H07_0$ Failed to Reject.
$H08_0$	Repo rate changes do not have a stronger impact on banking firms than automobile firms.	Tested in Model 3 & 4 (Tables 7 and 8)	$H08_0$ Failed to Reject for P/E. $H08_0$ Rejected for EV/EBITDA.

$H09_0$	Banking and automobile firms do not exhibit significantly different valuation behaviour over the period 2016–2024.	Tested in Model 3 & 4 (Tables 7 and 8)	$H09_0$ Rejected.
$H10_0$	There is no significant difference in valuation multiples between banking and automobile firms.	Tested in Figures 2, 3, 4 and 5.	$H10_0$ Failed to Reject for P/E. $H10_0$ Rejected for EV/EBITDA.

Table 9: Summary of Hypothesis Testing Results

Note: Rejected denotes the rejection of the Null Hypothesis at conventional significance levels based on model specific p values. For hypotheses involving both valuation measures, decisions are reported separately for P/E and EV/EBITDA where statistical outcomes differ.

4.8. Summary of Empirical Findings

The chapter has explored how monetary policy and key macroeconomic shocks affect equity-valuation multiples of Indian banking and automobile companies during the years between 2016 and 2024. Based on both earnings-based and enterprise-value-based value estimations, the empirical study offers a complete evaluation of the effect that various policy regimes and shocks have on firm valuations on a sense-by-sense and cross-sectional.

The results of the baseline regression are that responses of valuation multiples differ significantly between measures. With the case of P/E ratios, regular modification in the policy repo rate does not have a statistically significant influence. Nonetheless, macroeconomic shocks, especially the COVID-19 shock and extended macroeconomic tightening, are linked to substantial changes down in earnings-based valuations. This implies that P/E ratios are more vulnerable to long-term uncertainty and restrictive policy regimes rather than to small changes in the interest rates.

Conversely, EV/EBITDA exhibits a statistically significant sensitivity to interest rate changes once sectoral interaction effects are incorporated, while baseline effects remain weak. The fact that the correlation between the repo rate and the EV/EBITDA is negative means that the value of the enterprise-value based on the valuations adjust to the fluctuations in the financing costs

as well as the discount rate. Nonperiodic macroeconomic events such as demonetisation do not exhibit statistically significant effects on EV/EBITDA, while the COVID-19 shock shows a significant adverse impact in the enterprise-value-based interaction framework.

Introduction of interaction terms brings out sectoral asymmetry in valuation responses. In the case of P/E ratios, the ability to add the interaction effect of the sector has a significant explanatory advantage to the model. The findings reveal that the decrease in P/E ratios in banking companies is considerably higher compared to the situation in automobile companies during the agglomeration of monetary restrictions. This observation highlights the increased sensitivity of valuations in the banking-sector to extended restrictive monetary policy, given that the sector relies on credit expansion, interest margins and regulation.

In the case of EV/EBITDA, the effects of interaction also increase the performance of the models and demonstrate an alternative trend of sectoral differentiation. Although automobile firms show a high negative value of the EV/EBITDA in response to the cohort in the repo rate, the interaction establishment shows that the effect is much lower in the case of banking firms. This is an indicator that the enterprise value of banks is comparably immune to the rise in interest rates in relation to the automobile companies, and this is possibly because of variations in capital structure, ratio of revenues, and power of pricing.

The collective implication of the empirical results is that valuation reactions to monetary policy and macroeconomic shocks are not consistent regardless of the valuation indicator and by industry. Multiples based on earnings are more susceptible to periods of uncertainty and tightening, in particular, to the banking industry whereas multiples based on the enterprise value are more sensitive to interest-rate changes, in the automobile industry. Such findings indicate the need to account both the sectoral features and valuation techniques when evaluating the effect of macroeconomic policy on equity markets. These current findings are also supported by Welch t -test, which indicates that, even though the price-to-earnings ratios (P/E) do not vary significantly across industry sectors, the enterprise-value to earnings-before-interest, taxes, depreciation and amortisation ratios (EV/EBITDA) show that there is a statistically notable difference between industry sectors.

5. Chapter 5. Discussion and Implications

5.1. Empirical Results Discussion

The chapter is an interpretation and contextualisation of the presented empirical results in Chapter 4 in connection with existing literature regarding the monetary-policy transmission, valuation multiples, and the heterogeneity of sectors. The topic of discussion will focus on the explanation of why valuation multiples respond differently to the macroeconomic shocks and monetary policy to different sectors and valuation scales.

A. Monetary Policy and Valuation Multiples

At the level of standard policy repo rate alteration, the outcome of the regression at the baseline suggests the nonexistence of a statistically significant impact of regular shifts in the policy repo rate on either of the two indicators, i.e. the P/E ratios or the EV/EBITDA. However, depending on the directionality of the estimated coefficients, we can deduce different sensitivity to valuation which only achieve statistical significance when sectoral interaction effects are incorporated. This deviation highlights a significant difference between earnings and enterprise value-based valuation measures. P/E ratios are particularly susceptible to the volatility of the short-term earnings and the investor mood, which could avoid the immediate impact of an increase or drop in the policy rates. In comparison, the EV/EBITDA accounts for firm level operating cash flows and capital structure, which makes it more receptive to changes in discount rates and costs of financing.

These results are consistent with earlier data that has suggested that monetary policy affects asset prices based on discount-type of processes and not direct response to changes in earnings. (BERNANKE & KUTTNER, 2005) and (Rigobon & Sack, 2004) record that the effect of interest-rate changes on equity valuation is through the expected cash-flow discounting which has a more rigid correlation with measures of enterprise value, including EV/EBITDA. As such, (Gode & Ohlson, 2000) also point out that P/E multiples are less predictive to interest-rate fluctuations due to their sensitivity to accounting profits.

B. Effects of the Macroeconomic Shocks

The empirical findings indicate that major macroeconomic shocks have a significant impact on the value multiples especially P/E ratios. The COVID-19 dummy has a statistically significant negative effect on P/E ratios in the base model, which implies that increased levels of

uncertainty, earnings disturbance, and risk aversion because of the pandemic reduced earnings-based valuations. The same result can be found in (Behera et al., 2023) and (Mohanty & Mishra, 2024), who report negative stock-market performance in response to uncertainty related to the pandemic in emerging markets, including the case of India.

However, the statistical significance of long-term consequences of demonetisation in terms of price-to-earnings ratios and enterprise-value-to-EBITDA is not significant. On the other hand, the COVID -19 shock indicates a statistically significant negative effect on both earnings-based valuations and enterprise-value-based valuations when sectoral interactions are taken into account. This fact speaks in favour of the idea that despite the fact that demonetisation created a market shock in the short term, its effects on the valuation multiples of firms were not long-term in the cases of large, publicly listed firms. (Anoop et al., 2018) and (Upadhyay & Suvarna, 2018) also report similar findings, as the only effects of demonetisation, according to them, were mostly short-term and industry-specific.

Even factors that have been corrected by adjusting to other macroeconomic shocks reduce the P/E ratios statistically significantly with prolonged periods of monetary tightening. This brings out the significance of the policy regime change and not individual rate changes. Long tightening periods indicate an upward reflection of increased costs of borrowing and sluggish credit development as well as enhancement of macroeconomic danger and therefore, they are reflected in equity valuation. This fact supports the results by (Tomar & Kesharwani, 2022), who have reverse and regime-specific effects of monetary policy on Indian stock-market sectors.

5.2. The Effects of Sectors of Difference and Interaction

One of the most important contributions of the given research is its clear analysis of the sectoral asymmetry based on the model of interaction. The outcomes prove that the effects of monetary policy and macroeconomic shocks are not universal in all sectors.

In the case of P/E ratios, the interaction effects show that the drop in valuation in the banking firms is much high in times of monetary tightening compared to automobile firms. This increased sensitivity may be explained by the fact that the banking industry depends on the rise of credit, net interest rates, and the situation with the control, which is directly influenced by the tight monetary policy. According to (Ghosh, 2019) and (Sengupta, 2014), in India, the

transmission of monetary-policy works especially effectively in the banking industry, which is why the empiric-based findings of this study are justified.

The results of the interaction are different in the case of EV/EBITDA. The reaction of automobile firms to the rise in the repo rate is more significant whether in a negative sense whereas banking firms seem to be more resistant. This implies that the uncertainty of enterprise-value based valuation of automobile companies is also highly sensitive to financing cost, demand patterns and capital expenditures. Conversely, it is possible that banks can partially counter any increase in interest rates by reducing margin and repricing their balance sheets. These results are supported by industry-specific studies that point out that transmission through monetary policies differs across industries (Tomar & Kesharwani, 2022) (Ghosh, 2019).

The findings of the Welch t-test also support sectoral asymmetries, as the findings of the test reflect independently, intersectoral discrepancies in valuations. Even though there is no statistically significant difference between banking and automobile firms in terms of earnings-based valuation (P/E ratios), there is a strong and statistically significant difference across sectors in terms of enterprise-value based valuation (EV/EBITDA). This means that the findings are in line with the conclusion that sectoral heterogeneity in valuation is stronger at enterprise-value rather than earnings-level.

5.3. Implications for Investors and Market Participants

The results of this paper have various implications on the investors and managers of the portfolio. To begin with, being dependent on one valuation more than one can lead to hiding significant dynamics in the responses of firms to macroeconomic conditions. Multiples based on earnings, like P/E, are more susceptible to the long-run uncertainty and restrictive regimes especially in the banking industry. Conversely, the enterprise-value-based measures give better indications of interest-rate sensitivity particularly on the capital-intensive industries like the automobiles industry.

Second, sectoral asymmetry suggests that changes to monetary policy can shift relative valuations this reduces sector to sector. Sector-rotation strategies can be helpful to investors who can reweight portfolios depending on the expected monetary-policy regimes. In tightening cycles, the valuation of the banking-sector could be exposed to a higher downside risk in

earnings-related terms (as well as automobile-sector enterprise valuations to an increased sensitivity to increased financing costs).

5.4. Policy Implications

On policy front, the results of this paper would highlight the significance of acknowledging heterogeneous transmission of monetary policy within sectors. Whereas, the policy rates are the same throughout the economy, their effects on valuation vary depending on the structure of sector, the intensity of capital market and the composition of revenues. These asymmetries might require policy makers to consider the overall financial-market effects of periods of tightening (or easing) cycles.

5.5. Linkage to Existing Literature

On the whole, the results of this paper are widely related to the previous research describing the impacts of the monetary policy and macroeconomic shocks in determining asset prices (BERNANKE & KUTTNER, 2005) (Rigobon & Sack, 2004). Meanwhile, the study builds upon existing literature because it is based on valuation multiples as opposed to returns, a multi-shock model, and it explicitly investigates the effects of sector interactions in the situation of the Indian environment. By so doing, it helps towards a more refined choice on how the monetary policy and macroeconomics conditions affect the valuation of equity in emergent market environments.

6. Chapter 6. Conclusion

6.1. Conclusion

This paper examines how moves by the monetary policy and relevant macroeconomic shocks influence equity valuation multiples of a chosen sample of Indian banking and automobile companies during the time period of 2016-2024. The research provides a medium- to long-term view on the interaction between policy regimes and structural shocks on the valuation dynamics of the firm in the different sectors, since it focuses on valuation metrics including Price to earnings (P/E) and Enterprise Value to Earnings Before Interest Tax Depreciation and Amortization (EV/EBITDA).

In line with empirical evidence, the sensitivity of equity valuation to monetary policy is not uniform, but it is simultaneously influenced by valuation multiple used and the specifics of the industry of firms. Valuation measures that are sensitive to long durations of monetary tightening and increased macroeconomic uncertainty include earnings-based measures, especially P/E ratios. Conversely, being an enterprise-value based measure, EV/EBITDA exhibits greater sensitivity to interest-rate changes once sectoral interaction effects are incorporated, and thus captures the impact of financing costs and discounting mechanisms on firm valuation.

The discussion shows also strong valuation sectoral asymmetry. Banking companies are even more susceptible on P/E ratios on a long-term period of monetary tighten, according to their reliance on credit development, interest transmission, and balance sheets. However, auto companies are more taxing to interest-rate changes whenever valuations are determined based on EV/EBITDA, which underscores the importance of the capital intensity, price of financing, and demand factors in the enterprise value development. These results highlight the need to consider both organizational market structure as well as methodology of valuation in determining the impact of macroeconomic policy on the stock market.

These findings are further supported by the current analysis findings as reported by the cross-sector Welch t-test returns, which show no significant disparity in the earnings-based valuations (P/E), but that enterprise-value-based valuations (EV/EBITDA) show statistically significant patterns of the sector differentials.

On balance, the research proves that the effects of monetary policy measures and macroeconomic shocks on equity valuation occur in a variety of effects and by varying effects across industries and valuation measures. This combination of the baseline and interaction models provides a subtle insight into the valuation behaviour in Indian equity market and supports the argument of reasons why sector and metrics and analysis should be analysed separately when evaluating the effects of macroeconomic policy transmission to valuations of firms.

6.2. Key Contributions of the Study

The current paper provides a number of interesting contributions to the already existing research on the topic of the monetary-policy transmission and the topic of equity valuation. It first attempts to build upon previous studies by shifting the attention away from stock returns and volatility to valuation multiples, thus, anticipating the longer-term future of the market, as opposed to temporary fluctuations in prices. Second, it is multi-skilled research where several macroeconomic shocks are included in a single econometric model, which allows a comparative evaluation between different policy regimes and structural disturbances. Third, through the explicit modelling of the interaction effects of the sector, the study provides the evidence of skewed valuation reactions among both banking and automobile companies in the Indian setting. The contributions add value to the comprehension of the macroeconomic policy in influencing the equity valuation in emerging markets.

6.3. Limitations of the Study

In spite of contributions, the study has a number of limitations. The study is conducted with a limited sample of companies, which may inhibit the applicability of the research to the Indian equity market in general. Utilization of annual data can be a distortion of short-term a valuation-adaptations to monetary-policy declinations and macro-economic shocks.

Demonetization, COVID-19, and monetary tightening are examples of macroeconomic shocks represented using dummy variables which include regime effects but not entirely explain the severity or the timing of shocks at any given period. Furthermore, companies that experienced negative or very volatile profits such as Tata Motors were not included because of the inability to value them that can limit the coverage of the sector. The research does not directly include firm specific principals, like profitability, leverage, or growth principles, which can also result in valuation metrics. Lastly, the research is based on Ordinary Least Squares regression and

fails to consider any dynamic relationship and endogeneity. The Welch t -test, though allowing a resistant approach to comparative the means of the sectors under the condition of unequal variance, is a statistic textbook analytical tool that cannot represent any dynamic interaction between the two variables and does not reflect responses, which are constantly dynamically changing.

6.4. Scope for Future Research

This research can be expounded in one or more directions in future research. To increase generalizability, scholars could consider the sample that encompasses more areas and more companies as a unit of analysis. It might be more informative to use more high-frequency data, e.g., quarterly or monthly samples, to get closer to the understanding of the momentary in the market response to policy announcements. Future research might be able to incorporate firm-level financial factors to separate macro-economic impacts and firm-successful valuation factors. Other valuation measures such as price-to-book ratios or discounted cash flow–based indicators, should be considered. Also, more complex econometric models, such as panel vector autoregression, dynamic panel modelling, or regime-switching models can be utilized to represent the dynamics between the monetary policy and equity valuation.

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8. Appendices

Appendix A: List of Sample Firms

This appendix lists the companies that are included in the empirical study. The sample includes the leading publicly listed organizations in the banking and automobile industry in India, which are chosen based on the availability of data and its applicability to the market during the study period between 2016 and 2024.

Sector	Company Name
Banking	HDFC Bank Ltd.
Banking	ICICI Bank Ltd.
Banking	State Bank of India
Banking	Kotak Mahindra Bank Ltd.
Automobile	Maruti Suzuki India Ltd.
Automobile	Mahindra & Mahindra Ltd.
Automobile	Bajaj Auto Ltd.

Table 10: Sample Firms Included in the Study

Note: Tata Motors was not included in the sample due to the repeated negative operating profits of several years, which made the valuation multiples types of price-earnings (P/E) and enterprise value to earnings before interest, taxes, depreciation, and amortization (EV/EBITDA) invalid, as an analytical tool.

Appendix B: Variable Definitions and Coding

This appendix outlines the variables that were used in the empirical analysis that included their definitions, measurement procedures, coding schemes, and the sources of data.

Variable Name	Description	Measurement / Coding	Data Source
P/E Ratio	Price-to-Earnings ratio of the firm	Market price per share divided by earnings per share - Annual Values	(*Screener*, n.d.)
EV/EBITDA	Enterprise Value to EBITDA ratio	Enterprise value divided by EBITDA - Annual Values	(*Screener*, n.d.)
Repo Rate	Policy repo rate set by the Reserve Bank of India	Annual average repo rate (%)	(*Reserve Bank of India*, n.d.)
Demonetisation Dummy	Dummy variable capturing demonetisation period	1 for 2016, 0 Otherwise	Author's construction
COVID-19 Dummy	Dummy variable capturing COVID-19 shock period	1 for 2020 and 2021, 0 Otherwise	Author's construction
Monetary Tightening Dummy	Dummy variable capturing sustained monetary tightening phase	1 for 2022–2024, 0 Otherwise	Author's construction

Sector Dummy	Sector classification of firms	1 for Banking firms, 0 for Automobile firms	Author's classification

Table 11: Description of Variables

Notes:

- Valuation multiples are also calculated based on publicly issued financial data.
- Dummy variables have been developed in such a way that they represent macroeconomic regimes and not short-term market responses.

Appendix C: Regression Output

The appendix below provides the SPSS results that detailed the regression models estimated in the research study. The outputs are availed in reference and transparency to supplement the empirical findings in Chapter 4.

Variables Entered/Removed[a]

Model	Variables Entered	Variables Removed	Method
1	Monetary_Tightening_Dummy, Repo_Rate, Demonetisation_Dummy, COVID19_Dummy[b]	.	Enter

a. Dependent Variable: P/E

b. All requested variables entered.

Model Summary[b]

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Change Statistics		
					R Square Change	F Change	df1
1	.322[a]	.103	.039	10.931956169	.103	1.616	4

Model Summary[b]

Model	Change Statistics		Durbin-Watson
	df2	Sig. F Change	
1	56	.183	.955

a. Predictors: (Constant), Monetary_Tightening_Dummy, Repo_Rate, Demonetisation_Dummy, COVID19_Dummy

b. Dependent Variable: P/E

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	772.493	4	193.123	1.616	.183[b]
	Residual	6692.429	56	119.508		
	Total	7464.923	60			

a. Dependent Variable: P/E

b. Predictors: (Constant), Monetary_Tightening_Dummy, Repo_Rate, Demonetisation_Dummy, COVID19_Dummy

Figure 6: C1 – Part 1 - Regression Output – Model 1 (P/E as Dependent Variable)

Coefficients[a]

Model		Unstandardized Coefficients B	Unstandardized Coefficients Std. Error	Standardized Coefficients Beta	t	Sig.
1	(Constant)	47.180	12.415		3.800	<.001
	Repo_Rate	-2.879	1.996	-.282	-1.442	.155
	Demonetisation_Dummy	-4.448	5.009	-.128	-.888	.378
	COVID19_Dummy	-11.935	5.393	-.454	-2.213	.031
	Monetary_Tightening_Dummy	-7.434	3.564	-.319	-2.086	.042

Coefficients[a]

Model		Collinearity Statistics Tolerance	Collinearity Statistics VIF
1	(Constant)		
	Repo_Rate	.420	2.379
	Demonetisation_Dummy	.769	1.301
	COVID19_Dummy	.381	2.625
	Monetary_Tightening_Dummy	.683	1.464

a. Dependent Variable: P/E

Collinearity Diagnostics[a]

Model	Dimension	Eigenvalue	Condition Index	Variance Proportions (Constant)	Variance Proportions Repo_Rate	Variance Proportions Demonetisation_Dummy
1	1	2.735	1.000	.00	.00	.02
	2	1.009	1.646	.00	.00	.23
	3	1.000	1.654	.00	.00	.30
	4	.249	3.315	.01	.01	.41
	5	.007	19.931	.99	.99	.04

Collinearity Diagnostics[a]

Model	Dimension	Variance Proportions COVID19_Dummy	Variance Proportions Monetary_Tightening_Dummy
1	1	.01	.03
	2	.18	.01
	3	.04	.20
	4	.21	.66
	5	.57	.10

Figure 7: C1 – Part 2 - Regression Output – Model 1 (P/E as Dependent Variable)

Variables Entered/Removed[a]

Model	Variables Entered	Variables Removed	Method
1	Monetary_Tightening_Dummy, Repo_Rate, Demonetisation_Dummy, COVID19_Dummy[b]	.	Enter

a. Dependent Variable: EV/EBITDA

b. All requested variables entered.

Model Summary[b]

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Change Statistics		
					R Square Change	F Change	df1
1	.325[a]	.106	.044	4.8442031553	.106	1.711	4

Model Summary[b]

Model	Change Statistics		Durbin-Watson
	df2	Sig. F Change	
1	58	.160	.744

a. Predictors: (Constant), Monetary_Tightening_Dummy, Repo_Rate, Demonetisation_Dummy, COVID19_Dummy

b. Dependent Variable: EV/EBITDA

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	160.631	4	40.158	1.711	.160[b]
	Residual	1361.046	58	23.466		
	Total	1521.677	62			

a. Dependent Variable: EV/EBITDA

b. Predictors: (Constant), Monetary_Tightening_Dummy, Repo_Rate, Demonetisation_Dummy, COVID19_Dummy

Figure 8: C2 – Part 1 – Regression Output – Model 2 (EV/EBITDA as Dependent Variable)

Coefficients[a]

Model		Unstandardized Coefficients B	Unstandardized Coefficients Std. Error	Standardized Coefficients Beta	t	Sig.
1	(Constant)	27.817	5.482		5.075	<.001
	Repo_Rate	-1.648	.884	-.358	-1.864	.067
	Demonetisation_Dummy	-1.306	2.195	-.084	-.595	.554
	COVID19_Dummy	-3.487	2.359	-.295	-1.478	.145
	Monetary_Tightening_Dummy	.642	1.540	.062	.417	.678

Coefficients[a]

Model		Collinearity Statistics Tolerance	Collinearity Statistics VIF
1	(Constant)		
	Repo_Rate	.419	2.389
	Demonetisation_Dummy	.783	1.277
	COVID19_Dummy	.387	2.583
	Monetary_Tightening_Dummy	.707	1.414

a. Dependent Variable: EV/EBITDA

Collinearity Diagnostics[a]

Model	Dimension	Eigenvalue	Condition Index	Variance Proportions (Constant)	Variance Proportions Repo_Rate	Variance Proportions Demonetisation_Dummy
1	1	2.718	1.000	.00	.00	.02
	2	1.009	1.641	.00	.00	.24
	3	1.000	1.649	.00	.00	.31
	4	.267	3.192	.00	.01	.39
	5	.007	20.166	.99	.99	.05

Collinearity Diagnostics[a]

Model	Dimension	Variance Proportions COVID19_Dummy	Variance Proportions Monetary_Tightening_Dummy
1	1	.01	.03
	2	.18	.02
	3	.04	.21
	4	.20	.65
	5	.57	.10

Figure 9: C2 – Part 2 – Regression Output – Model 2 (EV/EBITDA as Dependent Variable)

Model Summary[b]

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Change Statistics		
					R Square Change	F Change	df1
1	.473[a]	.224	.104	10.556513814	.224	1.873	8

Model Summary[b]

Model	Change Statistics		Durbin-Watson
	df2	Sig. F Change	
1	52	.084	.946

a. Predictors: (Constant), Demonet_Sector, Covid_Sector, Monetary_Sector, Repo_Rate, Repo_Sector, Monetary_Tightening_Dur Demonetisation_Dummy, COVID19_Dummy

b. Dependent Variable: P/E

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	1670.043	8	208.755	1.873	.084[b]
	Residual	5794.879	52	111.440		
	Total	7464.923	60			

a. Dependent Variable: P/E

b. Predictors: (Constant), Demonet_Sector, Covid_Sector, Monetary_Sector, Repo_Rate, Repo_Sector, Monetary_Tightening_Dummy, Demonetisation_Dummy, COVID19_Dummy

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	47.547	11.992		3.965	<.001
	Repo_Rate	-3.466	1.981	-.339	-1.750	.086
	Demonetisation_Dummy	3.115	7.245	.090	.430	.669
	COVID19_Dummy	-11.213	6.358	-.426	-1.764	.084
	Monetary_Tightening_Dummy	1.761	4.925	.076	.357	.722
	Repo_Sector	1.000	.780	.261	1.282	.206
	Covid_Sector	-1.789	6.576	-.055	-.272	.787
	Monetary_Sector	-16.575	6.420	-.596	-2.582	.013
	Demonet_Sector	-13.692	9.630	-.306	-1.422	.161

Coefficients[a]

Model		Collinearity Statistics	
		Tolerance	VIF
1	(Constant)		
	Repo_Rate	.398	2.514
	Demonetisation_Dummy	.343	2.919
	COVID19_Dummy	.256	3.912
	Monetary_Tightening_Dummy	.334	2.997
	Repo_Sector	.360	2.779
	Covid_Sector	.371	2.697
	Monetary_Sector	.280	3.565
	Demonet_Sector	.321	3.111

a. Dependent Variable: P/E

Figure 10: C3 – Part 1 – Regression Output – Model 3 (P/E with Interaction Terms)

Collinearity Diagnostics[a]

Model	Dimension	Eigenvalue	Condition Index	Variance Proportions (Constant)	Repo_Rate	Demonetisation _Dummy
1	1	3.991	1.000	.00	.00	.00
	2	1.768	1.503	.00	.00	.03
	3	1.756	1.508	.00	.00	.04
	4	.563	2.662	.00	.01	.00
	5	.367	3.296	.00	.00	.11
	6	.245	4.035	.00	.00	.19
	7	.244	4.043	.00	.00	.27
	8	.060	8.177	.01	.02	.33
	9	.007	24.242	.98	.97	.03

Collinearity Diagnostics[a]

Model	Dimension	Variance Proportions COVID19_Dummy	Monetary_Tightening_Dummy	Repo_Sector	Covid_Sector	Monetary_Sector
1	1	.00	.01	.01	.00	.01
	2	.04	.00	.00	.06	.00
	3	.01	.03	.00	.01	.03
	4	.00	.00	.05	.07	.09
	5	.05	.16	.17	.02	.02
	6	.27	.01	.00	.42	.01
	7	.05	.20	.00	.08	.20
	8	.25	.57	.76	.34	.62
	9	.34	.03	.02	.00	.01

Collinearity Diagnostics[a]

Model	Dimension	Variance ... Demonet_Sector
1	1	.00
	2	.03
	3	.04
	4	.05
	5	.01
	6	.21
	7	.27
	8	.39
	9	.00

a. Dependent Variable: P/E

Figure 11: C3 – Part 2 –Regression Output – Model 3 (P/E with Interaction Terms)

Model Summary[b]

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Change Statistics		
					R Square Change	F Change	df1
1	.582[a]	.339	.241	4.3159833376	.339	3.461	8

Model Summary[b]

Model	Change Statistics		Durbin-Watson
	df2	Sig. F Change	
1	54	.003	1.081

a. Predictors: (Constant), Demonet_Sector, Covid_Sector, Monetary_Sector, Repo_Rate, Repo_Sector, Monetary_Tightening_Dum Demonetisation_Dummy, COVID19_Dummy

b. Dependent Variable: EV/EBITDA

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	515.781	8	64.473	3.461	.003[b]
	Residual	1005.896	54	18.628		
	Total	1521.677	62			

a. Dependent Variable: EV/EBITDA

b. Predictors: (Constant), Demonet_Sector, Covid_Sector, Monetary_Sector, Repo_Rate, Repo_Sector, Monetary_Tightening_Dummy, Demonetisation_Dummy, COVID19_Dummy

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	27.817	4.884		5.696	<.001
	Repo_Rate	-2.106	.807	-.457	-2.610	.012
	Demonetisation_Dummy	.732	2.962	.047	.247	.806
	COVID19_Dummy	-5.622	2.595	-.476	-2.166	.035
	Monetary_Tightening_Dummy	1.462	2.013	.140	.726	.471
	Repo_Sector	.801	.307	.471	2.610	.012
	Covid_Sector	3.735	2.664	.253	1.402	.167
	Monetary_Sector	-1.435	2.578	-.115	-.556	.580
	Demonet_Sector	-3.567	3.894	-.177	-.916	.364

Coefficients[a]

Model		Collinearity Statistics	
		Tolerance	VIF
1	(Constant)		
	Repo_Rate	.399	2.508
	Demonetisation_Dummy	.341	2.931
	COVID19_Dummy	.254	3.937
	Monetary_Tightening_Dummy	.328	3.046
	Repo_Sector	.377	2.655
	Covid_Sector	.376	2.660
	Monetary_Sector	.288	3.467
	Demonet_Sector	.328	3.049

a. Dependent Variable: EV/EBITDA

Figure 12: C4 – Part 1 – Regression Output – Model 4 (EV/EBITDA with Interaction Terms)

Collinearity Diagnostics[a]

Model	Dimension	Eigenvalue	Condition Index	Variance Proportions (Constant)	Repo_Rate	Demonetisation _Dummy
1	1	3.963	1.000	.00	.00	.00
	2	1.767	1.498	.00	.00	.03
	3	1.756	1.502	.00	.00	.04
	4	.556	2.670	.00	.01	.00
	5	.402	3.138	.00	.00	.10
	6	.245	4.021	.00	.00	.19
	7	.244	4.030	.00	.00	.27
	8	.060	8.149	.01	.02	.33
	9	.007	24.469	.98	.97	.03

Collinearity Diagnostics[a]

Model	Dimension	Variance Proportions COVID19_Dummy	Monetary_Tightening_Dummy	Repo_Sector	Covid_Sector	Monetary_Sector
1	1	.00	.01	.01	.00	.01
	2	.04	.00	.00	.06	.00
	3	.01	.03	.00	.01	.03
	4	.00	.00	.03	.08	.11
	5	.05	.15	.18	.01	.01
	6	.26	.01	.00	.43	.01
	7	.05	.20	.00	.08	.21
	8	.25	.58	.77	.33	.61
	9	.34	.03	.01	.00	.00

Collinearity Diagnostics[a]

Model	Dimension	Variance ... Demonet_Sector
1	1	.00
	2	.03
	3	.04
	4	.06
	5	.01
	6	.21
	7	.28
	8	.38
	9	.00

a. Dependent Variable: EV/EBITDA

Figure 13: C4 – Part 2 – Regression Output – Model 4 (EV/EBITDA with Interaction Terms)

Appendix D: Data Sources

This appendix identifies the secondary data sources that will be used in the work.

Data Type	Description	Source
Policy Interest Rate	Monetary policy stance is used in form of an annual policy repo rate.	(*Reserve Bank of India*, n.d.)
Valuation Multiples	Firm-level P/E and EV/EBITDA ratios.	(*Screener*, n.d.)
Firm Financial Information	Market capitalization, earnings, enterprise value components.	(*Screener*, n.d.)

Table 12: Data Sources

Notes:

- The data utilized in the research are all of the secondary type, and are in the open.
- Cross-checking Valuation multiples was done to check consistency among years.
- The research is based on the annual data to maintain comparability between firms and years.

Annexure-VI

Undertaking from the PG student (s) while submitting his /her final Dissertation report to his/her respective institute

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